

Unit 17

CARBONYL COMPOUNDS

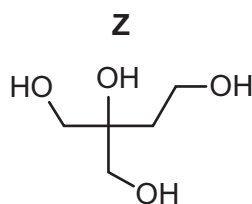
50 items · 191 marks

Syllabus sections in this unit

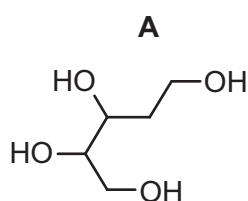
17.1 Aldehydes and ketones

Unit 17: Carbonyl compounds**1. M/J 26/P21/Q4/b(ii)**

- 4 (a) Compound **Z** is oxidised to produce organic compound **W**, $\text{C}_5\text{H}_6\text{O}_7$, by heating under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$.

**Figure 4.1**

- (b) Compound **A** is a structural isomer of **Z**.

**Figure 4.2**

- (ii) Suggest why Fehling's reagent does **not** react with **A** but does react with aldehyde functional groups.

[1]

2. M/J 26/P22/Q6/c(iii)

6 (a) Figure 6.1 shows the structure of compound **P**.

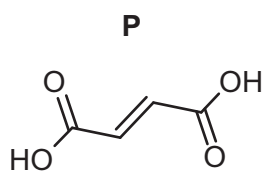


Figure 6.1

(c) An unlabelled sample contains one of compounds **P**, **R** or **S**.

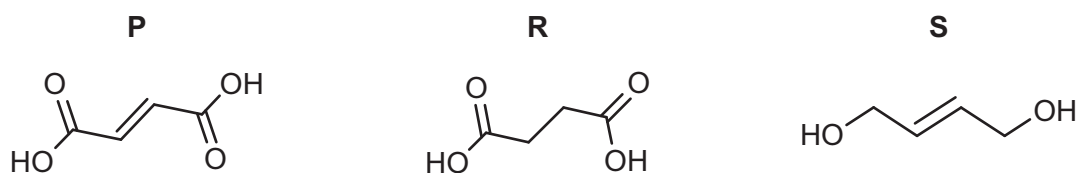


Figure 6.4

The mass spectrum of the sample is shown in Figure 6.5.

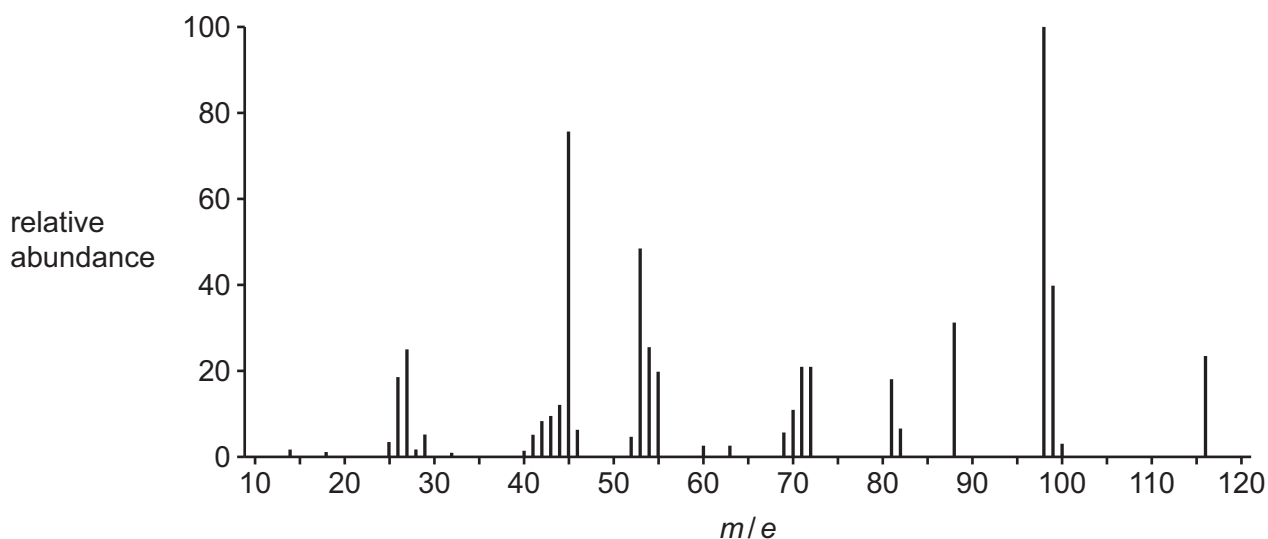


Figure 6.5

(iii) Suggest **one** reagent that is used to distinguish between samples of **R** and **S**. Include appropriate observations.

[2]

3. M/J 26/P23/Q4/a(iii)

4 The structure of compound **V** is shown in Figure 4.1.

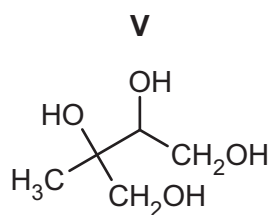


Figure 4.1

- (iii) Suggest why Fehling's reagent does **not** react with **V** but does react with aldehyde functional groups.

[1]

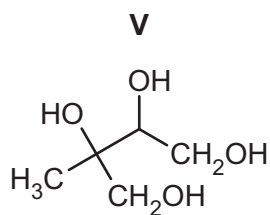


Figure 4.1

4. M/J 25/P21/Q6/a,b,c(i,ii)

- 6** Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.

(a) State the functional group in ethanal, propanal and propanone.

[1]

(b) State a reagent and the relevant observation that confirm that **F**, **G** and **H** have the same functional group.

reagent

observation

[2]

(c) Separate samples of **F**, **G** and **H** are each tested with Tollens' reagent and with alkaline $I_2(aq)$. The observations are shown in Table 6.1.

Table 6.1

	Tollens' reagent	alkaline $I_2(aq)$
F	no observable change	pale yellow precipitate
G	silver mirror	pale yellow precipitate
H	silver mirror	no precipitate

(i) Use Table 6.1 to name the organic compounds in bottles **F**, **G** and **H**.

F =

G =

H =

[2]

(ii) Identify the yellow precipitate produced when alkaline $I_2(aq)$ is added to separate samples of **F** and **G**.

[1]

5. MAR 25/P22/Q4/b(i)

4 Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

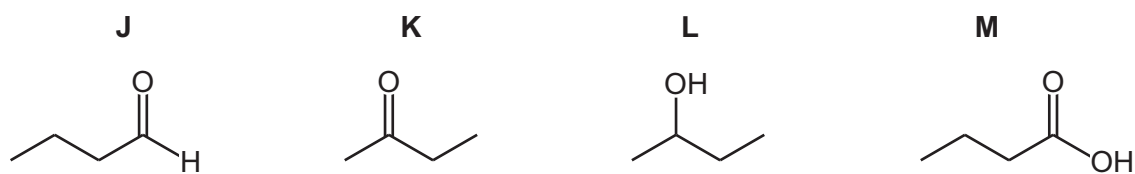


Fig. 4.1

(b) **K** reacts with HCN in the presence of a KCN catalyst, forming **N**.

(i) Complete Fig. 4.2 to show the mechanism for this reaction.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

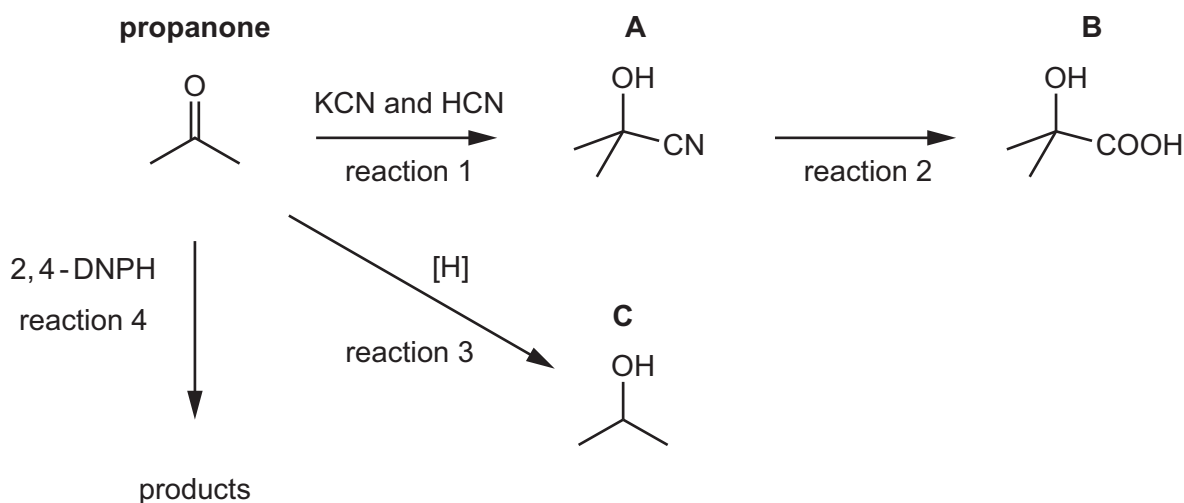


Fig. 4.2

[4]

6. O/N 24/P21/Q4/a(i),c,d,e

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

- (a)** Reaction 1 is a nucleophilic addition reaction.
- (i)** Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.
- Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.

**Fig. 4.2**

- (c)** Reaction 3 is a reduction reaction.
- (i)** Construct an equation to represent reaction 3.
- Use $[\text{H}]$ to represent one atom of hydrogen from the reducing agent.

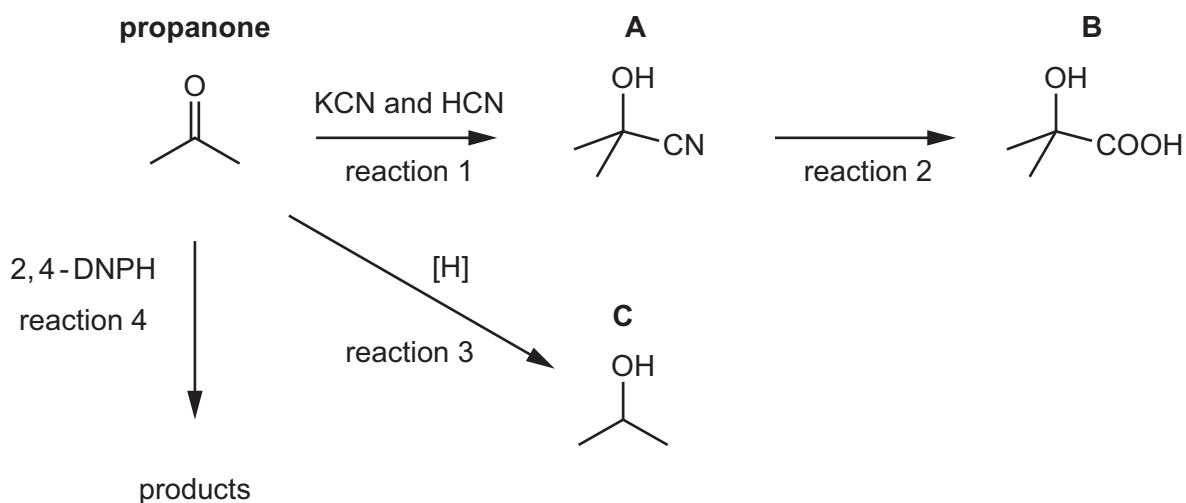
- (ii)** Name **C**.

- (d)** State what is observed in reaction 4.

- (e)** Explain why Fehling's reagent does **not** react with propanone.

7. O/N 24/P23/Q4/a(i),c,d,e

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- Use $[\text{H}]$ to represent one atom of hydrogen from the reducing agent.

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- (d)** State what is observed in reaction 4.

- (e)** Explain why Fehling's reagent does **not** react with propanone.

8. M/J 24/P21/Q6/b(i),c(ii)

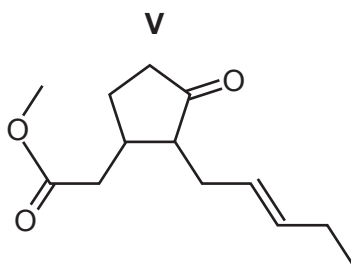
6 (a) **V** shows stereoisomerism.

Fig. 6.1

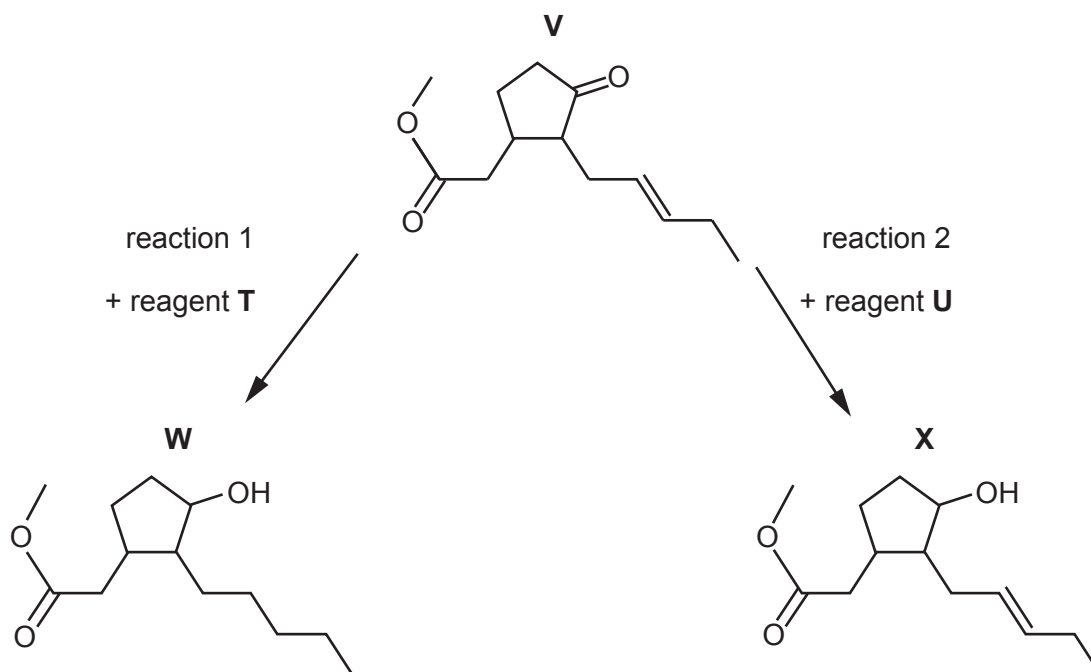
(b) Fig. 6.2 shows two reactions involving **V**.

Fig. 6.2

(i) Identify the role of reagent **T** for each functional group that reacts in reaction 1.

[1]

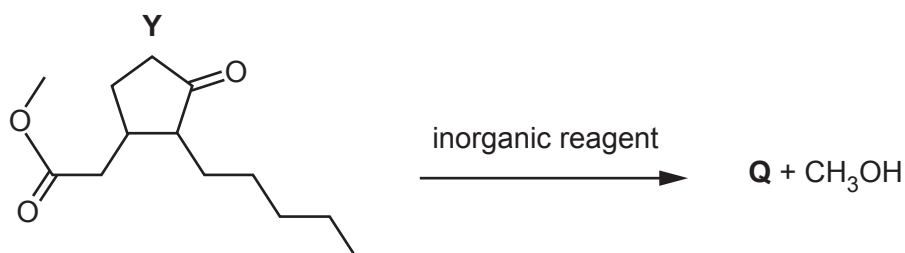
(c) Both functional groups in one molecule of **Y** react with an inorganic reagent to form one molecule of **Q** and one molecule of methanol, CH_3OH , as shown in Fig. 6.3.

Fig. 6.3

(ii) **Q** contains **only** hydroxyl functional groups.

Complete Table 6.1 to show the observations that occur when 2,4-dinitrophenylhydrazine (2,4-DNPH reagent) is added to separate samples of **Y** and **Q**.

Table 6.1

	observation on addition of 2,4-DNPH reagent
Y	
Q	

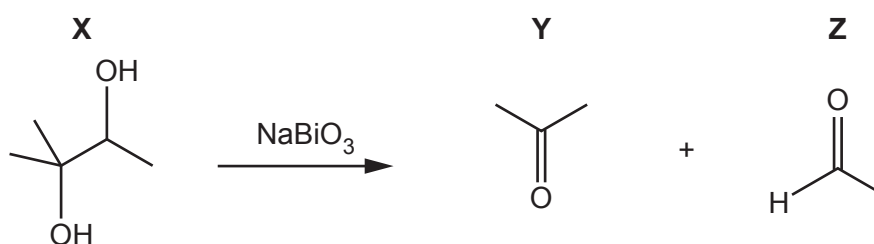
[1]

9. MAR 24/P22/Q1/d(iii,iv)

1 Bismuth is an element in Group 15 of the Periodic Table.

(d) NaBiO_3 is an oxidising agent with similar properties to KMnO_4 .

Fig. 1.1 shows an example of the use of NaBiO_3 as an oxidising agent.

**Fig. 1.1**

(iii) Complete Table 1.1 to show what is observed when compounds **Y** and **Z** react separately with the named reagents.

Table 1.1

reagent	observation with Y	observation with Z
$\text{Na}_2\text{CO}_3(\text{aq})$	no reaction	
alkaline $\text{I}_2(\text{aq})$		
2,4-dinitrophenylhydrazine (2,4-DNPH)		
Tollens' reagent		

[4]

(iv) Construct an equation for the reaction of **Z** with NaBH_4 .

Use [H] to represent an atom of hydrogen from the reducing agent.

[1]

10. M/J 23/P21/Q4/a

4 **V** is a colourless liquid.

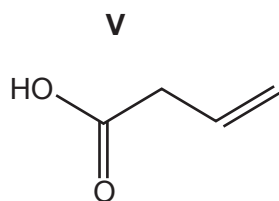
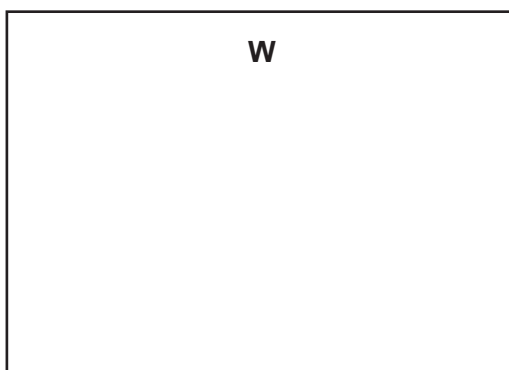


Fig. 4.1

(a) **V** reacts with an excess of LiAlH_4 to form **W**.

(i) Draw the structure of **W** in the box.



[1]

(ii) Identify the role of LiAlH_4 in the reaction with **V**.

[1]

11. M/J 23/P22/Q5/b

(b) **A** and **B** are structural isomers with molecular formula $C_5H_{10}O$.

They are both straight-chained molecules with only one functional group.

Table 5.1 describes observations when separate samples of **A** and **B** are added to different reagents.

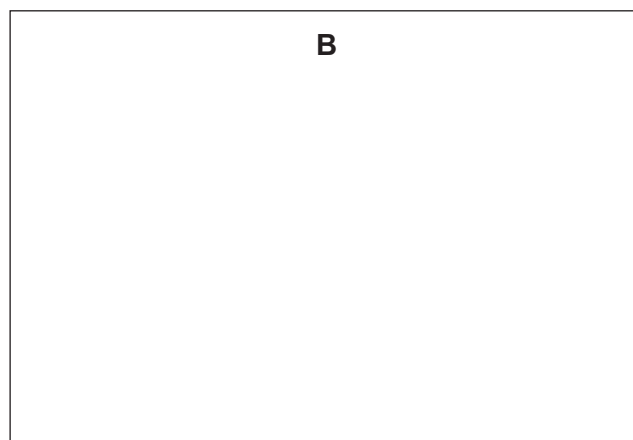
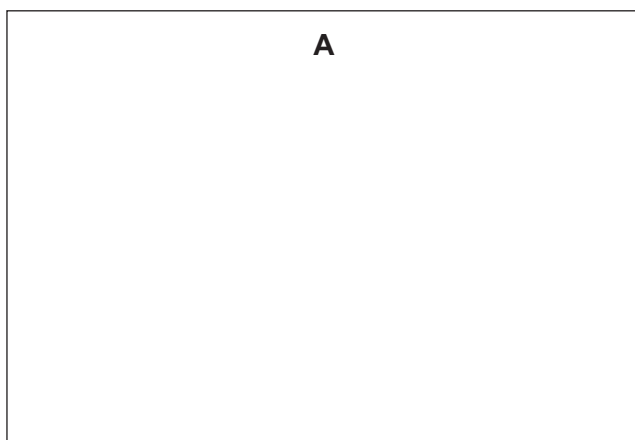
Table 5.1

reagent	A	B
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	orange precipitate appears	orange precipitate appears
Tollens' reagent	silver mirror appears	no reaction
alkaline $I_2(aq)$	no reaction	no reaction

(i) Name the functional group present in both **A** and **B**.

[1]

(ii) Draw the structures of **A** and **B** in the boxes.



[2]

12. M/J 23/P23/Q5/b,d

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.

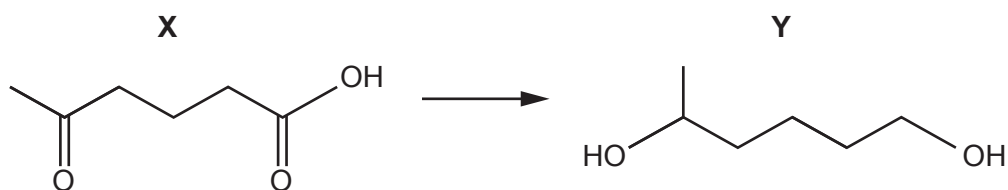


Fig. 5.1

(b) The formation of **Y** from **X** requires the addition of a suitable reducing agent.

(i) Construct an equation using molecular formulae and [H] for the reaction in Fig. 5.1. Use [H] to represent one atom of hydrogen from the reducing agent.

[1]

(ii) Identify a suitable non-gaseous reducing agent for the formation of **Y** from **X**.

[1]

(d) Complete Table 5.2 with the expected observations that occur when the reagents shown are added to separate solutions of **X** and **Y**. Do **not** refer to temperature changes in your answer.

Table 5.2

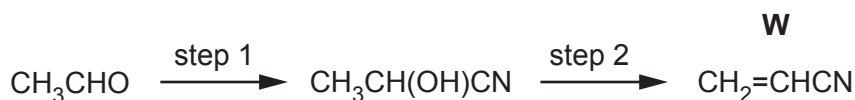
reagent	observation on addition to X	observation on addition to Y
aqueous sodium carbonate		
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)		
alkaline aqueous iodine		

[3]

13. M/J 23/P23/Q6/b(i,ii)

6 Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

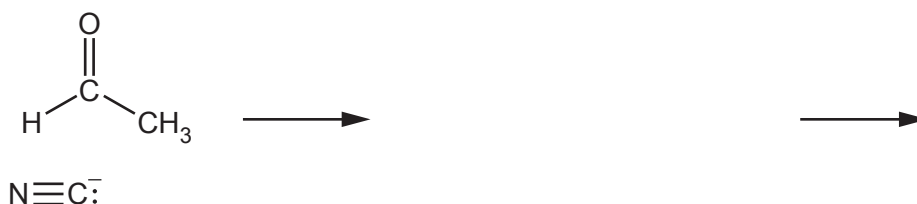
- (b)** CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN .

**Fig. 6.1**

- (i)** Name the mechanism for the reaction in step 1 in Fig. 6.1.

[1]

- (ii)** Complete Fig. 6.2 to show the mechanism for the reaction in step 1.
Include all products, charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 6.2**

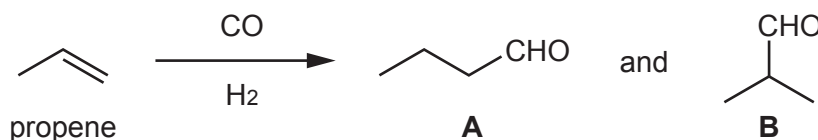
[3]

14. MAR 23/P22/Q3/b

also Unit 13 (1 mark); this unit 8 marks

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2 .

**Fig. 3.1**

- (b)** **A** and **B** show reactions typical of aliphatic aldehydes.

- (i)** **A** undergoes a nucleophilic addition reaction with a mixture of HCN and KCN , forming compound **C**.

Complete the diagram to show the mechanism for this reaction.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

Draw the structure of the organic intermediate.



[4]

(ii) Table 3.1 shows information about three experiments involving **B**.

Complete Table 3.1.

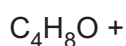
Table 3.1

experiment	reagents	observation with B
1		solution turns from orange to green
2		a silver mirror forms on the sides of the reaction vessel
3	$\text{Br}_2(\text{aq})$	

[3]

(iii) **B**, $\text{C}_4\text{H}_8\text{O}$, is oxidised by acidified potassium manganate(VII).

Complete the equation for this reaction. Use [O] to represent one atom of oxygen from the oxidising agent.



[1]

(iv) **C** is a chiral molecule.

Circle any chiral centres in the structure of **C** shown in Fig. 3.2.

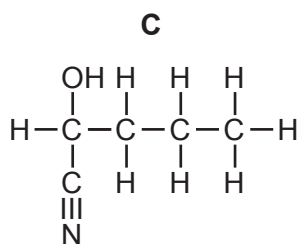


Fig. 3.2

[1]

15. O/N 22/P21/Q5/a(ii,iii)

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.

Equation 1 shows an example of the formation of a lactone.

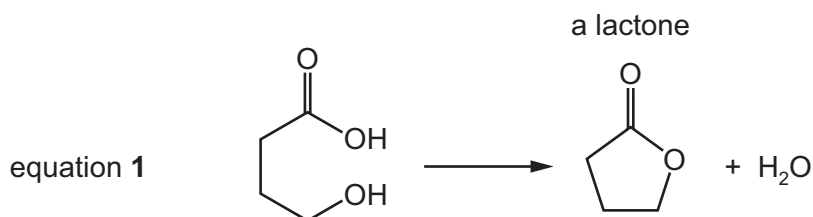


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

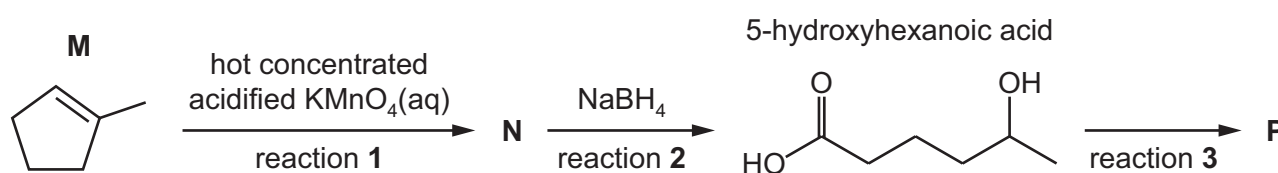


Fig. 5.1

- (ii)** **N** is reduced by NaBH_4 to form 5-hydroxyhexanoic acid in reaction 2.

Construct an equation for reaction 2 using molecular formulae.

In the equation, use $[\text{H}]$ to represent one atom of hydrogen from the reducing agent.

[1]

- (iii)** Reaction 2 is a nucleophilic addition.

Suggest why reaction 2 creates a mixture of two organic compounds.

[2]

16. O/N 22/P23/Q5/a(ii,iii)

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.

Equation 1 shows an example of the formation of a lactone.

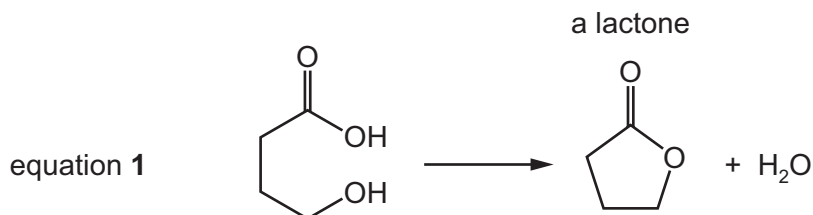
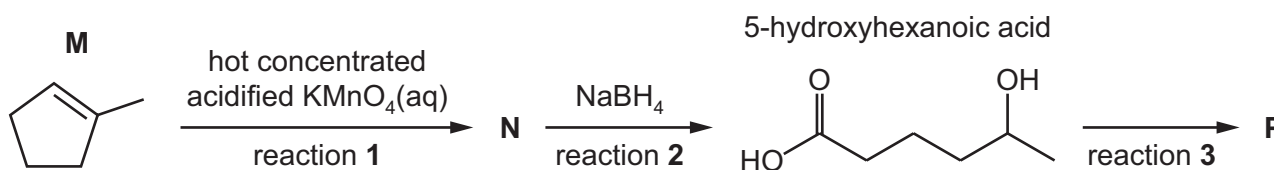


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.



- (ii)** **N** is reduced by NaBH_4 to form 5-hydroxyhexanoic acid in reaction 2.

Construct an equation for reaction 2 using molecular formulae.

In the equation, use $[\text{H}]$ to represent one atom of hydrogen from the reducing agent.

[1]

- (iii)** Reaction 2 is a nucleophilic addition.

Suggest why reaction 2 creates a mixture of two organic compounds.

[2]

17. M/J 22/P21/Q3/c(i)

3 Liquids that contain molecules of **T** smell like lemons.

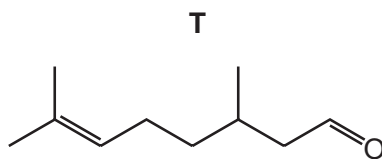


Fig. 3.1

(c) Fig. 3.2 shows two reactions of **T**.

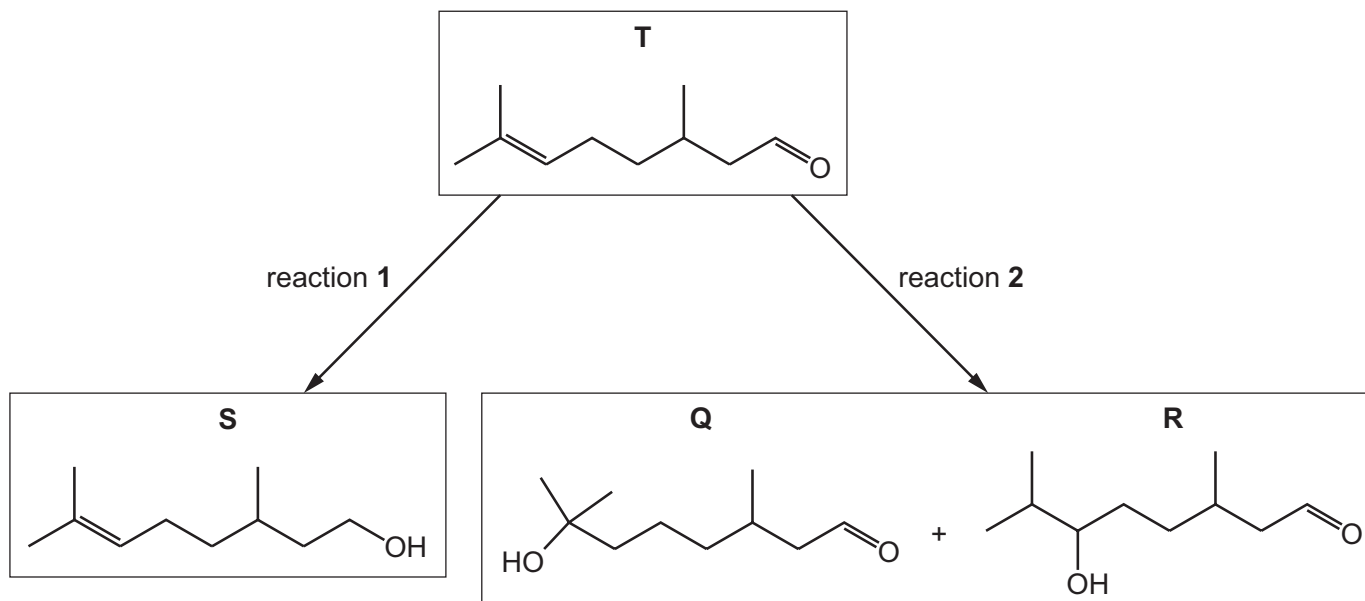


Fig. 3.2

(i) Identify a suitable reagent for reaction 1.

[1]

18. M/J 22/P21/Q4/c(i,ii)

4 Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

(c) **W**, **X** and **Y** have the same molecular formula, $C_5H_{10}O$.

W, **X** and **Y** are added separately to different reagents. Observations for these reactions are described in Table 4.1.

Table 4.1

	+ 2,4-dinitrophenylhydrazine	+ alkaline $I_2(aq)$	+ Fehling's reagent and warm
W	orange precipitate seen	no change	orange-red precipitate seen
X	orange precipitate seen	yellow precipitate seen	no change
Y	orange precipitate seen		

(i) **W**, **X** and **Y** each contain a common functional group.

Name the functional group that is present in all three compounds.

[1]

(ii) State the formula of the yellow precipitate produced when **X** is added to alkaline $I_2(aq)$.

[1]

19. M/J 22/P23/Q3/c

3 **G** belongs to a group of compounds called ethers.

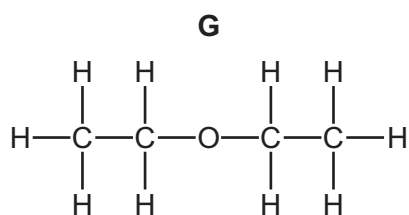


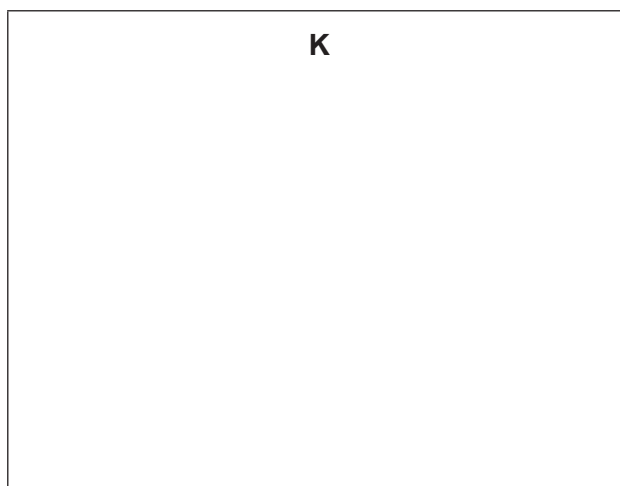
Fig. 3.1

(c) **K** has molecular formula $\text{C}_3\text{H}_6\text{O}$.

When **K** is added to 2,4-dinitrophenylhydrazine, an orange precipitate forms.

When **K** is warmed with Tollens' reagent, a silver mirror forms.

Draw the displayed formula of **K**.



[2]

20. MAR 22/P22/Q4/a(i)

4 Compounds **J** and **K** are found in plant oils.

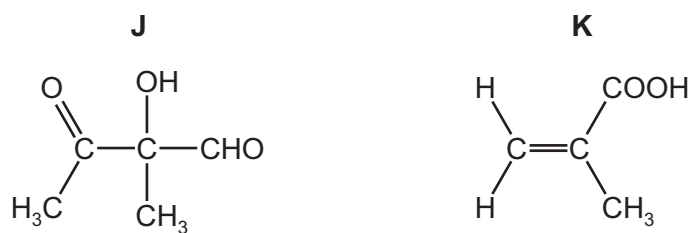


Fig. 4.1

(a) (i) Complete Table 4.1 to state what you would **observe** when **J** reacts with the reagents listed.

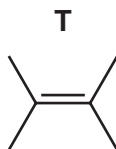
Table 4.1

reagent	observation with J
2,4-dinitrophenylhydrazine (2,4-DNPH)	
Tollens' reagent	
sodium metal	

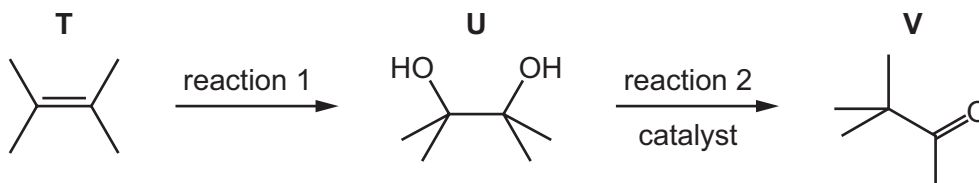
[3]

21. O/N 21/P22/Q3/d(ii),e(i,iii)

3 Compound **T** is an isomer of C_6H_{12} .



(d) A reaction scheme starting with **T** is shown. Reaction 2 occurs in the presence of a catalyst; knowledge of the mechanism for this reaction is not required.

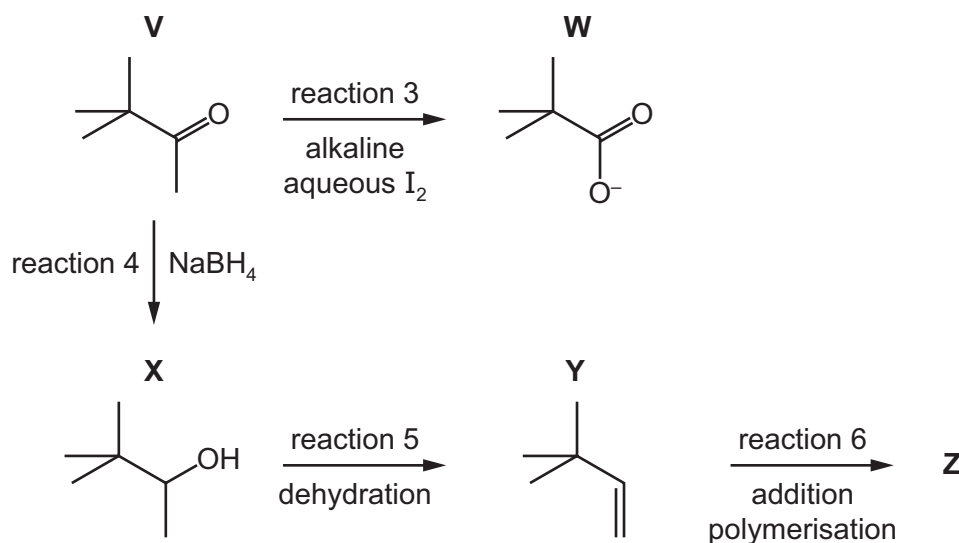


(ii) State and explain how 2,4-dinitrophenylhydrazine (2,4-DNPH) can be used to detect the presence of **V** as a product of reaction 2.

[2]

(e) **V** is used in a wide range of organic reactions.

Some reactions of **V** are shown.



(i) **V** and **W** are colourless and soluble in water.

State what you would observe in reaction 3.

[1]

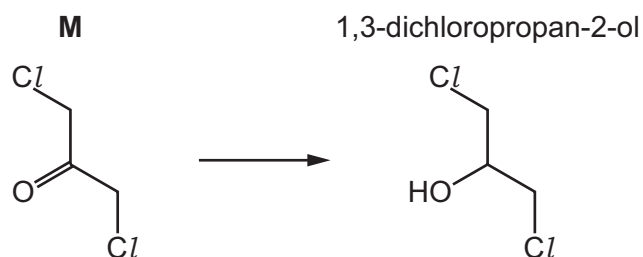
(iii) Construct an equation for reaction 4.

Use [H] in the equation to represent an atom of hydrogen from $NaBH_4$.

$C_6H_{12}O + \dots\dots\dots$ [1]

22. M/J 21/P23/Q4/a(ii,iii)

4 (a) 1,3-dichloropropan-2-ol can be made by reacting **M**.



(ii) Name the functional group present in **M** that changes during this reaction.

[1]

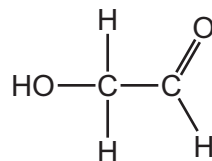
(iii) State a suitable reagent for this reaction.

[1]

23. MAR 21/P22/Q4/b,d(i,ii)

- 4** Hydroxyethanal, HOCH_2CHO , has been observed in dust clouds near the centre of our galaxy.

hydroxyethanal



- (b)** Hydroxyethanal reacts separately with 2,4-dinitrophenylhydrazine (2,4-DNPH) and with Tollens' reagent.

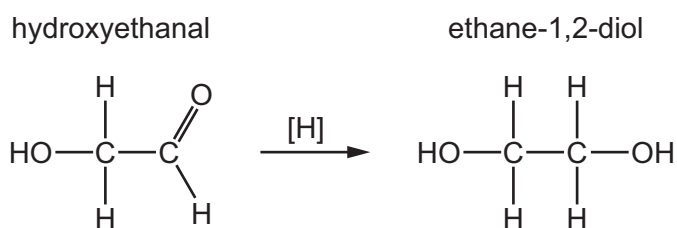
State what you would observe in each reaction.

reaction with 2,4-DNPH

reaction with Tollens' reagent

[2]

- (d)** Hydroxyethanal can be reduced to ethane-1,2-diol, $(\text{CH}_2\text{OH})_2$, as shown.



- (i)** Write an equation for the reduction of hydroxyethanal to $(\text{CH}_2\text{OH})_2$.

Use $[\text{H}]$ to represent an atom of hydrogen from the reducing agent.

[1]

- (ii)** Identify a reagent for this reduction reaction.

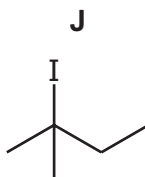
[1]

24. O/N 20/P21/Q4/c

also Unit 13 (2 marks), Unit 15 (2 marks), Unit 14 (1 mark); this unit 3 marks

- 4** Iodine is used in many inorganic and organic reactions.

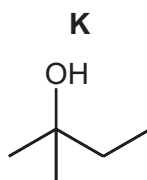
- (c)** **J** reacts with NaOH , forming different products dependent on the conditions used.



- (i)** Name **J**.

[1]

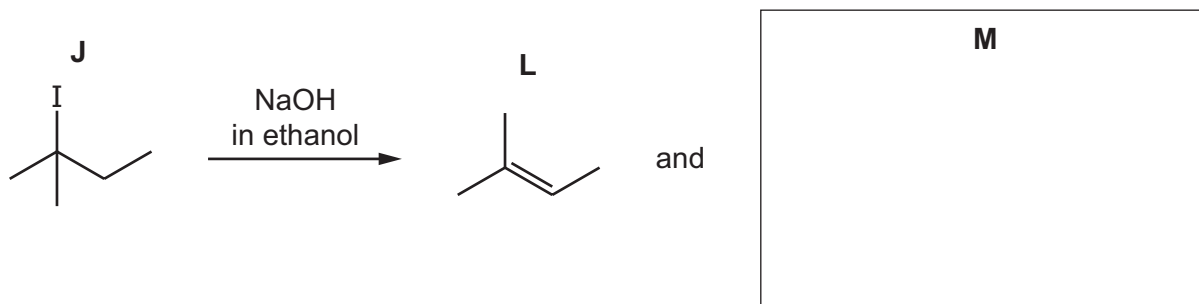
- (ii)** **J** reacts with NaOH(aq) to form **K**.



Fully name the mechanism of the reaction of **J** with NaOH(aq) to form **K**.

[1]

- (iii) **J** reacts with NaOH dissolved in ethanol to form a mixture of two alkenes, **L** and **M**. Alkene **L** is shown.



In the box provided, draw the structure of **M**.

[1]

- (iv) Explain why **L** does **not** show geometrical (cis-trans) isomerism.

[1]

- (v) **L** reacts with hot concentrated acidified $\text{KMnO}_4(\text{aq})$ to form propanone and one other organic product.

Identify the other organic product.

[1]

- (vi) Propanone reacts with excess alkaline aqueous iodine.

Complete and balance the equation for this reaction.



[2]

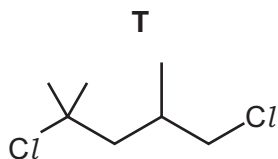
- (vii) State **one** observation that can be made in the reaction in (c)(vi).

[1]

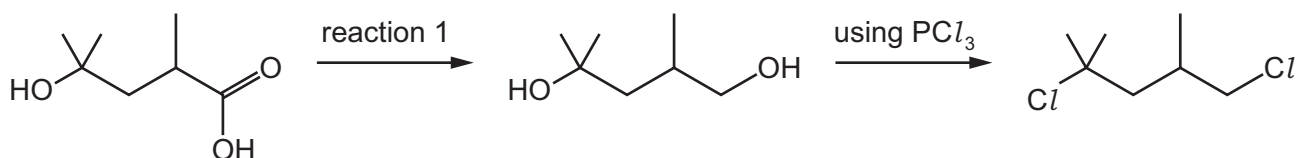
25. O/N 20/P22/Q3/c(i)

3 PCl_5 , PCl_3 and NCl_3 are halides of Group 15 elements.

(c) PCl_3 is used to convert alcohols to chloroalkanes, such as compound **T**.



A possible synthesis of **T** is shown.

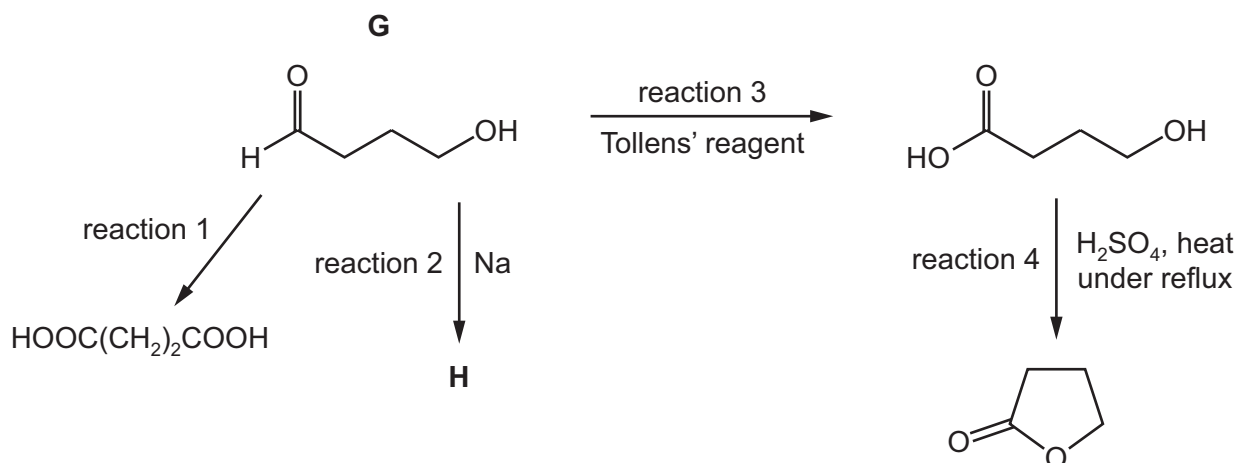


(i) Identify a reagent that could be used in reaction 1.

[1]

26. O/N 20/P22/Q4/a(iv),b(ii,iii,iv)

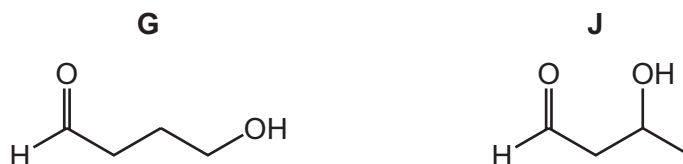
4 Some reactions of compound **G** are shown.



(iv) State what you would observe in reaction 3.

[1]

(b) **G** and **J** are structural isomers of each other.



(ii) Suggest **one** chemical test that can distinguish **G** from **J**. Give the result of the test with each compound.

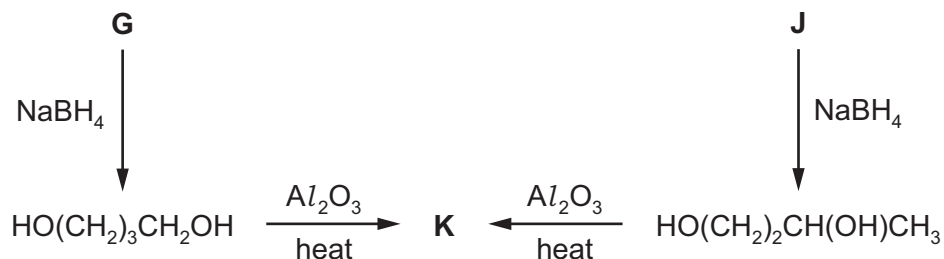
test

result with **G**

result with **J**

[2]

In the reaction schemes below, **G** and **J** are converted into organic compound **K**.



(iii) State the role of NaBH_4 in the reactions with **G** and **J**.

[1]

(iv) Identify the organic product **K**.

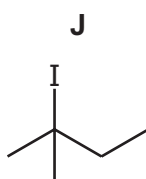
[1]

27. O/N 20/P23/Q4/c

also Unit 13 (2 marks), Unit 15 (2 marks), Unit 14 (1 mark); this unit 3 marks

4 Iodine is used in many inorganic and organic reactions.

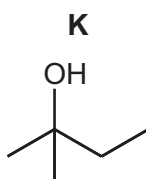
(c) **J** reacts with NaOH , forming different products dependent on the conditions used.



(i) Name **J**.

[1]

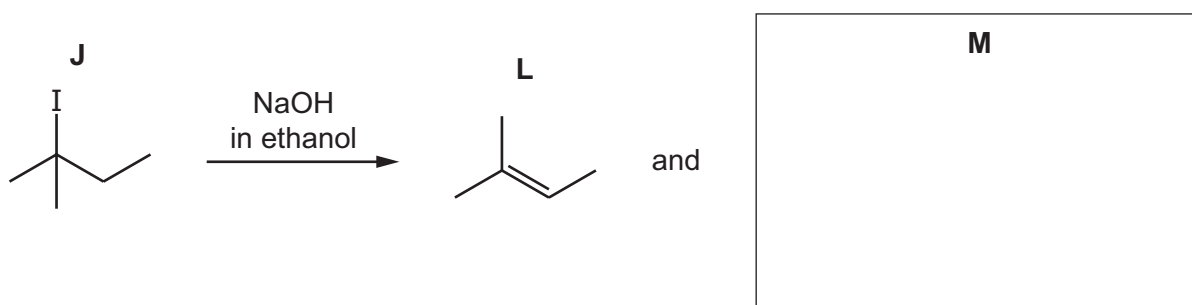
(ii) **J** reacts with NaOH(aq) to form **K**.



Fully name the mechanism of the reaction of **J** with NaOH(aq) to form **K**.

[1]

(iii) **J** reacts with NaOH dissolved in ethanol to form a mixture of two alkenes, **L** and **M**. Alkene **L** is shown.



In the box provided, draw the structure of **M**.

[1]

(iv) Explain why **L** does **not** show geometrical (cis-trans) isomerism.

[1]

(v) **L** reacts with hot concentrated acidified $\text{KMnO}_4(\text{aq})$ to form propanone and one other organic product.

Identify the other organic product.

[1]

(vi) Propanone reacts with excess alkaline aqueous iodine.

Complete and balance the equation for this reaction.



[2]

(vii) State **one** observation that can be made in the reaction in (c)(vi).

[1]

28. M/J 20/P21/Q5/c(i)

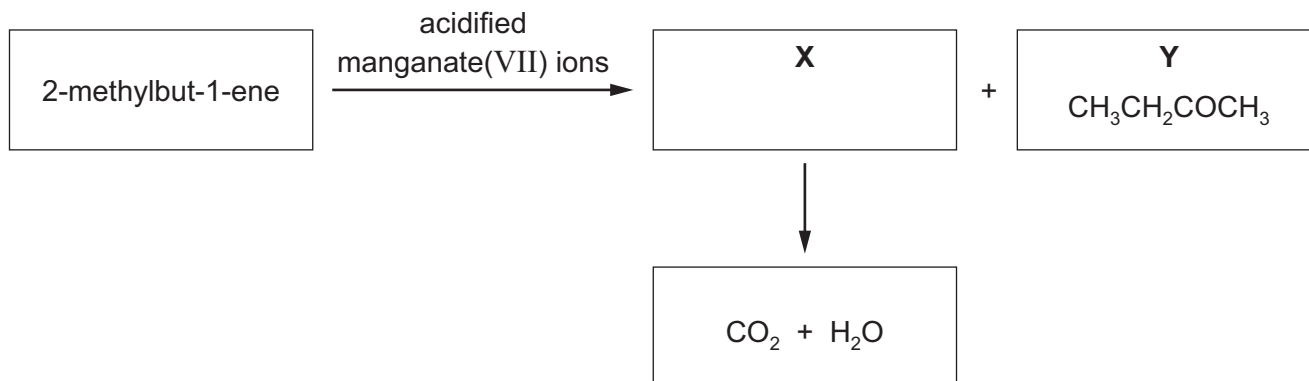
(c) (i) Name an organic functional group which reacts with a nucleophile in an addition reaction.

[1]

29. M/J 20/P21/Q6/d

- 6** 2-methylbut-1-ene reacts with acidified manganate(VII) ions, under specific conditions, to produce two organic compounds **X** and **Y**.

X immediately reacts with the acidified manganate(VII) ions to form carbon dioxide and water. **Y** has the structural formula $\text{CH}_3\text{CH}_2\text{COCH}_3$.

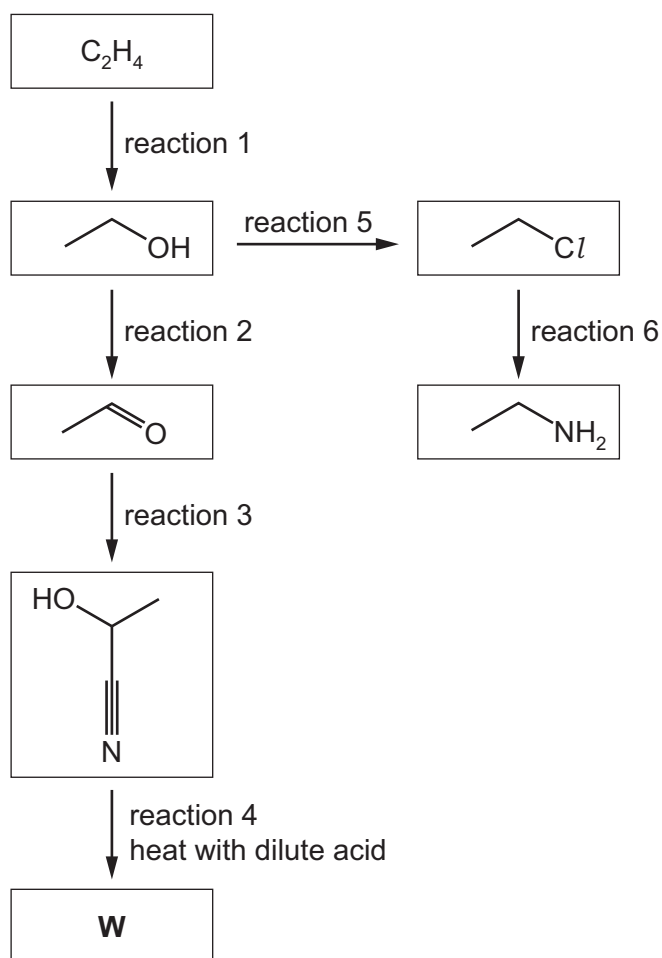


- (d)** Describe a chemical test and the expected observation(s) to confirm the presence of the carbonyl functional group in **Y**.

[2]

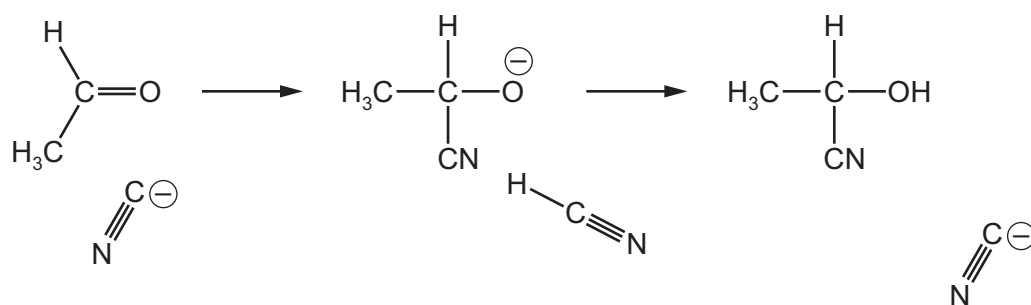
30. M/J 20/P22/Q5/b(i)

5 The reaction sequence shows how ethene, C_2H_4 , can be converted into other organic molecules.



(b) In reaction 3 the organic molecule reacts with HCN and a KCN catalyst.

(i) Complete the diagram to show the mechanism of the reaction occurring. Include all relevant dipoles, lone pairs and curly arrows in your answer.

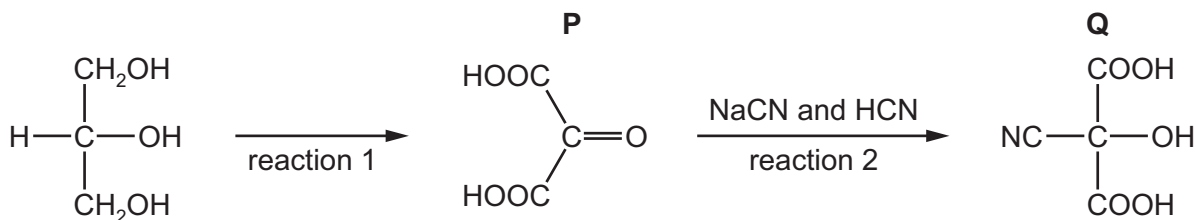


[3]

31. MAR 20/P22/Q3/a(ii,iii)

3 Glycerol, $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$, is widely used in the food industry and in pharmaceuticals.

(a) A series of reactions starting from glycerol is shown.



(ii) Name the reaction mechanism for reaction 2.

[1]

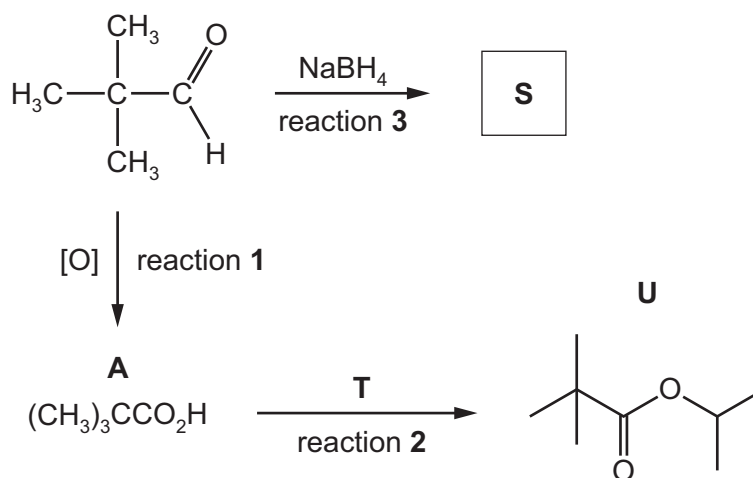
(iii) Give the observation you would make when 2,4-dinitrophenylhydrazine is added to **P**.

[1]

32. O/N 19/P21/Q4/b(i,iii),c(i,ii,vii,viii)

4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(b) Two reaction sequences are shown.



(i) Reaction 1 is an oxidation reaction.

Identify the reagent(s) and conditions for reaction 1.

[1]

(iii) Give the balanced equation for the reaction of $(\text{CH}_3)_3\text{CCHO}$ with NaBH_4 to form **S**.

Use $[\text{H}]$ to represent an atom of hydrogen provided by NaBH_4 .

[1]

(c) **X**, **Y** and **Z** are all isomers of $(\text{CH}_3)_3\text{CCHO}$.

A summary of some of the reactions and properties of **X**, **Y** and **Z** is shown in the table.

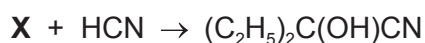
compound	observations with 2,4-DNPH	observations with Fehling's solution	principal absorptions in infra-red spectrum
X		no reaction	1715 cm ⁻¹
Y		red precipitate	1730 cm ⁻¹
Z	no reaction	no reaction	3200–3600 cm ⁻¹ 1630 cm ⁻¹ 1050 cm ⁻¹

(i) **X** and **Y** each contains a carbonyl group.

Complete the table with the expected observations for the reactions of **X** and **Y** with 2,4-DNPH. [1]

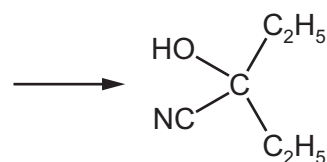
(ii) Identify the functional group present in **Y** that causes the recorded observation with Fehling's solution. [1]

(vii) **X** contains a carbonyl group. **X** reacts with HCN, in the presence of a small amount of NaCN, to form (C₂H₅)₂C(OH)CN as shown.



Draw the mechanism of the reaction of **X** with HCN.

- Draw the structure of **X** and the intermediate.
- Include all charges, partial charges, lone pairs and curly arrows.



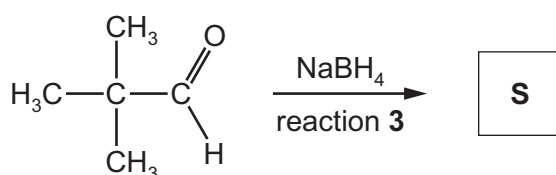
[3]

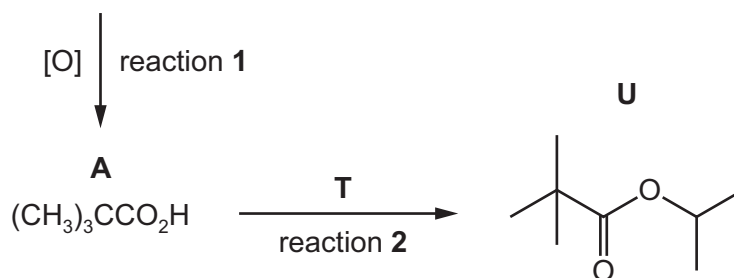
(viii) State the role of NaCN in the reaction in (c)(vii). [1]

33. O/N 19/P23/Q4/b(i,iii),c(i,ii,vii,viii)

4 (CH₃)₃CCHO is used in the synthesis of some antibiotics.

(b) Two reaction sequences are shown.





(i) Reaction 1 is an oxidation reaction.

Identify the reagent(s) and conditions for reaction 1.

[1]

(iii) Give the balanced equation for the reaction of $(\text{CH}_3)_3\text{CCHO}$ with NaBH_4 to form **S**.

Use [H] to represent an atom of hydrogen provided by NaBH_4 .

[1]

(c) **X**, **Y** and **Z** are all isomers of $(\text{CH}_3)_3\text{CCHO}$.

A summary of some of the reactions and properties of **X**, **Y** and **Z** is shown in the table.

compound	observations with 2,4-DNPH	observations with Fehling's solution	principal absorptions in infra-red spectrum
X		no reaction	1715 cm^{-1}
Y		red precipitate	1730 cm^{-1}
Z	no reaction	no reaction	$3200\text{--}3600\text{ cm}^{-1}$ 1630 cm^{-1} 1050 cm^{-1}

(i) **X** and **Y** each contains a carbonyl group.

Complete the table with the expected observations for the reactions of **X** and **Y** with 2,4-DNPH. [1]

(ii) Identify the functional group present in **Y** that causes the recorded observation with Fehling's solution.

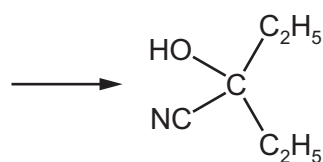
[1]

(vii) **X** contains a carbonyl group. **X** reacts with HCN , in the presence of a small amount of NaCN , to form $(\text{C}_2\text{H}_5)_2\text{C}(\text{OH})\text{CN}$ as shown.



Draw the mechanism of the reaction of **X** with HCN .

- Draw the structure of **X** and the intermediate.
- Include all charges, partial charges, lone pairs and curly arrows.



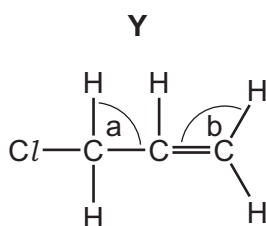
[3]

(viii) State the role of NaCN in the reaction in (c)(vii).

[1]

34. M/J 19/P21/Q4/d(i)

4 The structure of compound **Y** is shown.



(d) Samples of organic compounds, **A**, **B**, **C** and **D**, are placed in unlabelled bottles.



(i) Identify all of the compound(s), **A–D**, that contain a carbonyl group.

[1]

35. M/J 19/P21/Q5/a

5 Ethanal reacts with a mixture of HCN and NaCN to make 2-hydroxypropanenitrile, $\text{CH}_3\text{CH}(\text{OH})\text{CN}$.

The reaction mechanism is nucleophilic addition.

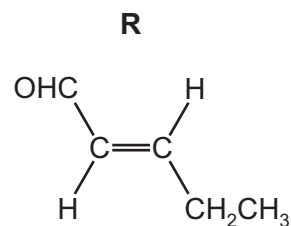
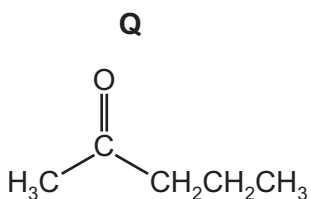
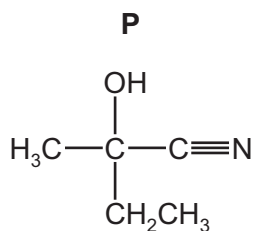
(a) Explain the meaning of the term *nucleophile* and identify the species which acts as the nucleophile during this reaction.

species acting as nucleophile

[2]

36. MAR 19/P22/Q3/b

3 **P**, **Q** and **R** all contain five carbon atoms.

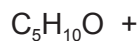


A student carries out several tests to distinguish between **P**, **Q** and **R**.

(b) **Q** is reduced by NaBH_4 .

Write an equation for the reaction of **Q** with NaBH_4 .

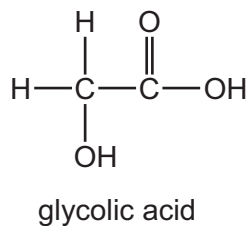
In your answer, use $[\text{H}]$ to represent NaBH_4 .



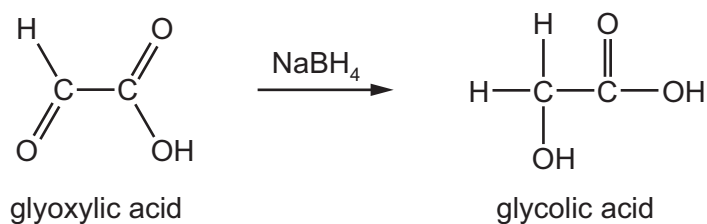
[1]

37. O/N 18/P21/Q4/c

4 The structure of glycolic acid is shown.



(c) Glycolic acid can also be made by reacting glyoxylic acid with NaBH_4 .



(i) State the role of NaBH_4 in this reaction.

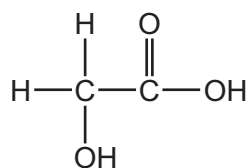
[1]

(ii) Write an equation for this reaction using molecular formulae.
Use $[\text{H}]$ to represent NaBH_4 .

[2]

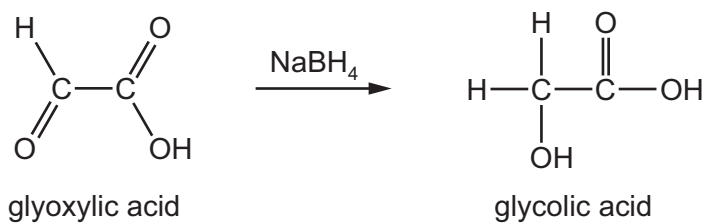
38. O/N 18/P23/Q4/c

4 The structure of glycolic acid is shown.



glycolic acid

(c) Glycolic acid can also be made by reacting glyoxylic acid with NaBH_4 .



(i) State the role of NaBH_4 in this reaction.

[1]

(ii) Write an equation for this reaction using molecular formulae.
Use [H] to represent NaBH_4 .

[2]

39. M/J 18/P22/Q4/a(i),c,d(i)

4 **W** is $\text{CH}_3\text{COCH}_2\text{CH}_3$.

(a) The reaction between **W** and alkaline aqueous iodine produces a yellow precipitate.

(i) Give the name of the compound formed as a yellow precipitate in this reaction.

[1]

(c) (i) Identify the type of reaction occurring in reaction 1.

[1]

(ii) Identify the reagent for reaction 1.

[1]

(d) Reaction 2 is carried out by adding a mixture of HCN and NaCN to **W**.
The product, **X**, is formed as a mixture of two isomers.

(i) Complete the mechanism for this reaction.

Include the structure of the intermediate formed and all necessary charges, dipoles, lone pairs and curly arrows.



[4]

40. M/J 17/P21/Q4/c,d(ii)

4 **P**, **Q** and **R** all have the molecular formula C_3H_6O . They are all structural isomers of each other.

P and **Q** each contain an oxygen atom bonded directly to a carbon atom that is sp^2 hybridised.
R contains an oxygen atom bonded directly to a carbon atom that is sp^3 hybridised.

(c) **P** and **Q** (C_3H_6O) both form an orange precipitate when reacted with 2,4-DNPH. Only **Q** produces a yellow precipitate when reacted with alkaline aqueous iodine.

(i) Name **P** and **Q**.

P

Q

[2]

(ii) Identify the yellow precipitate formed by the reaction of **Q** with alkaline aqueous iodine.

[1]

(d) **P** and **Q** each react with hydrogen cyanide to form a single product.

The product formed from **P** exists as a pair of optical isomers.

The product formed from **Q** does not exhibit optical isomerism.

(ii) Ethanal, CH_3CHO , also reacts with hydrogen cyanide. The product of this reaction is $CH_3CH(OH)CN$.

Draw the mechanism of this reaction.

Include all necessary charges, dipoles, lone pairs and curly arrows.

[3]

41. M/J 17/P22/Q4/c(i,ii)

(c) The isomers **F** and **G**, $C_5H_{10}O$, both form an orange precipitate when reacted with 2,4-DNPH.

F is unbranched and reacts with alkaline aqueous iodine to produce a yellow precipitate.

G does not react with alkaline aqueous iodine. It contains a chiral centre and produces a silver mirror when warmed with Tollens' reagent.

(i) Name the yellow precipitate produced by the reaction between **F** and alkaline aqueous iodine.

[1]

(ii) Give the structural formula of **F** and of **G**.

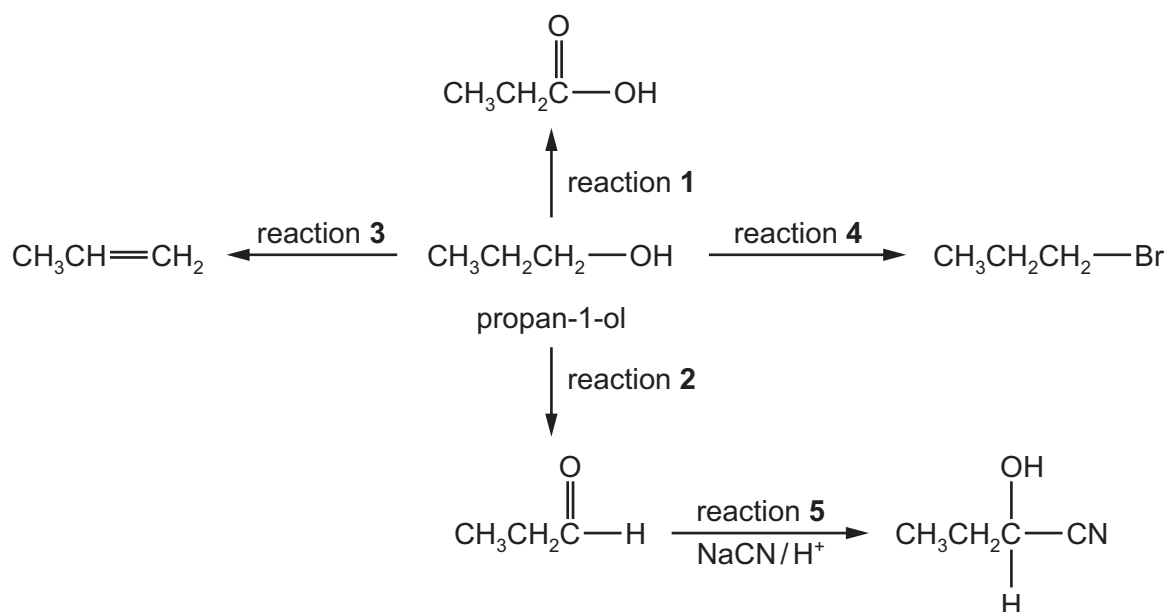
F

G

[2]

45. M/J 16/P22/Q5/c(i,iii)

5 A reaction sequence based on propan-1-ol is shown.



- (c) (i)** Complete the reaction mechanism for reaction **5**. Include all relevant lone pairs, curly arrows, charges and partial charges.



[4]

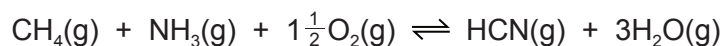
The product of reaction **5** exhibits stereoisomerism.

- (iii)** Suggest why a mixture of the two stereoisomers is formed by reaction **5**.

[2]

46. MAR 16/P22/Q3/d

- 3 Over one million tonnes of hydrogen cyanide, HCN, are produced each year using the Andrussow process. The overall equation for the reaction is shown.



- (d) The reaction of hydrogen cyanide with propanone is an important first step in many organic syntheses.

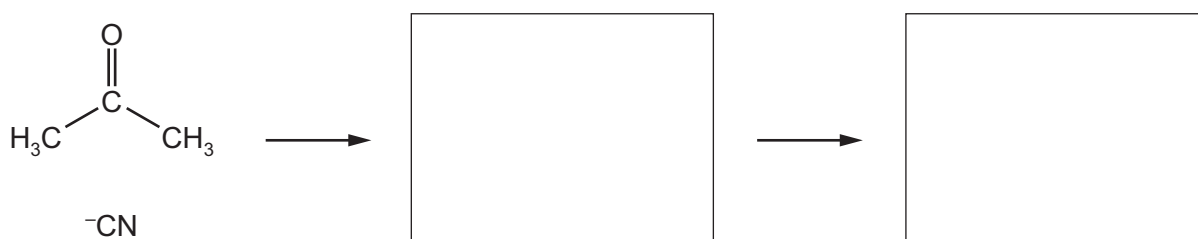
(i) Give the full name of the mechanism of this reaction.

[1]

- (ii) Complete the diagram to show the mechanism of the reaction of hydrogen cyanide with propanone.

Draw the structure of the intermediate and the product of the reaction.

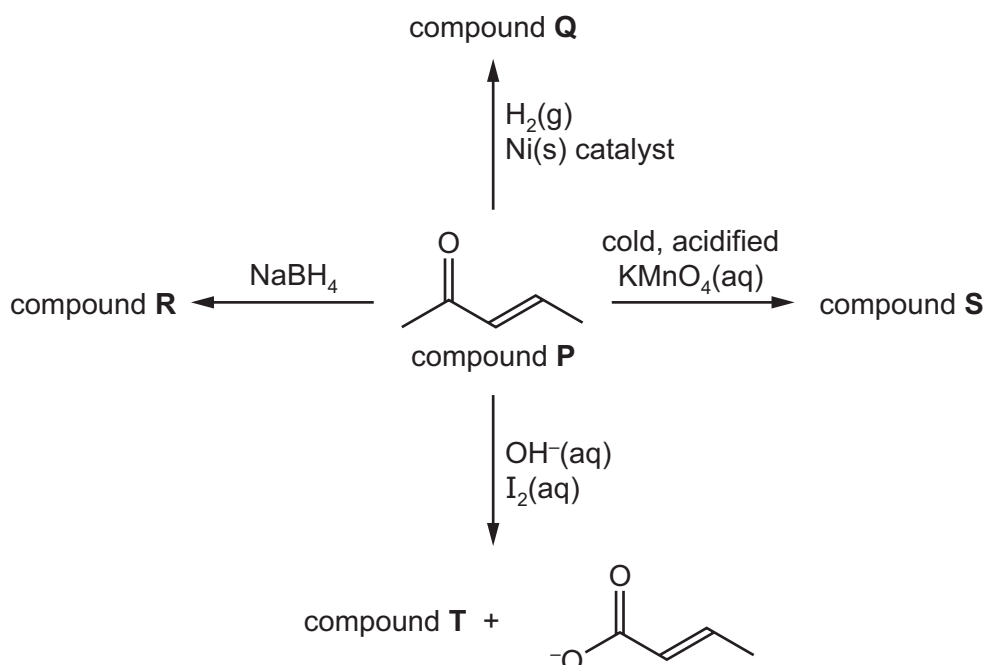
Include all relevant charges, partial charges, curly arrows and lone pairs.



[5]

47. MAR 16/P22/Q5/a(iii)

- 5 Some reactions of compound **P**, C₅H₈O, are shown.

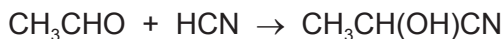


- (iii) What would you observe when compound **P** is reacted with 2,4-dinitrophenylhydrazine (2,4-DNPH)?

[1]

48. M/J 15/P21/Q3/b(ii),c

- 3** Ethanal reacts with hydrogen cyanide, in the presence of a small amount of NaCN, as shown.

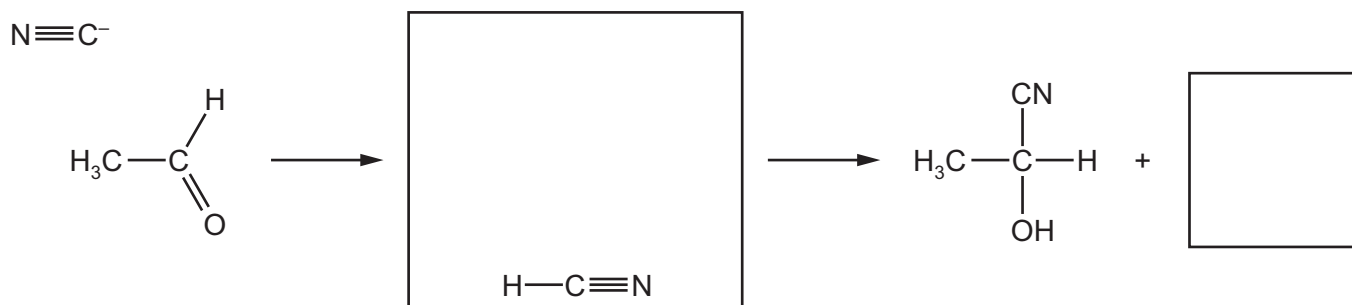


- (b)** The product of this reaction shows stereoisomerism as it contains a chiral centre. This reaction produces an equimolar mixture of two optical isomers.

- (ii)** Suggest why the two optical isomers are produced in equal amounts by this reaction.

[1]

- (c) (i)** Complete the diagram to show the mechanism of this reaction. Include all necessary charges, partial charges, lone pairs and curly arrows and show the structure of the intermediate.



[5]

- (ii)** With reference to your mechanism in **(i)**, explain the role of the NaCN in this reaction.

[1]

49. M/J 15/P22/Q4/b

- 4** There are seven structural isomers with the molecular formula $\text{C}_5\text{H}_{10}\text{O}$ that are carbonyl compounds. Four of these are aldehydes.

These four aldehydes, **A**, **B**, **C** and **D**, have the following properties.

- Aldehyde **A** has a straight chain while **B**, **C** and **D** are branched.
- Aldehyde **B** is the only one of the four isomers with a chiral centre and it exists as a pair of optical isomers.
- Aldehyde **C** has two methyl groups in its structure but **D** has three.

- (b) (i)** Describe a chemical test that would allow you to distinguish between any of the four isomers **A** to **D** and any of the other three structural isomers of $\text{C}_5\text{H}_{10}\text{O}$, that are carbonyl compounds.

In your answer you should describe any necessary reagents and conditions as well as explaining what you would **see** in each case.

[3]

- (ii)** Describe a test that would give the same result with all seven carbonyl isomers of $\text{C}_5\text{H}_{10}\text{O}$.

[2]

50. M/J 15/P23/Q3/a(i,iii),c

3 **A, B, C, D, E** and **F** are all structural isomers with the molecular formula C_4H_8O .

(a) **A, B** and **C** all give an orange precipitate when treated with 2,4-DNPH but only **A** and **B** give a brick-red precipitate when warmed with Fehling's solution.

(i) Draw the **skeletal** formulae of **A, B** and **C**.

A	B	C

[3]

(iii) State what you would **see** when a sample of **A** is warmed with Tollens' reagent.

[1]

(c) Another compound, **G**, C_3H_6O , contains the same functional group as **A**.

Give equations for the reactions of **G** with each of acidified potassium dichromate(VI) and sodium tetrahydridoborate, $NaBH_4$, using [O] or [H] as appropriate.

(i) reaction with acidified potassium dichromate(VI)



(ii) reaction with $NaBH_4$



Answers Section

1. M/J 26/P21/Q4/b(ii)

4(b)(ii)	Alcohols / Compound A are / is not oxidised by Fehling's reagent (Aldehydes are oxidised by Fehling's reagent) OR Aldehydes are oxidised by Fehling's reagent (Alcohols / Compound A are / is not oxidised by Fehling's reagent)	1 A for alcohol groups = hydroxy = hydroxyl = OH group R for alcohol groups = hydroxide / OH⁻ group A alcohol groups are not a strong enough reducing agent (to reduce Fehling's) A Fehling's reagent is not a strong enough oxidising agent to oxidise the alcohol groups A Fehling's is a strong enough oxidising agent to oxidise aldehydes A Fehling's is too mild to oxidise alcohols A Fehling's isn't reduced by compound A / alcohols A Fehling's is reduced by aldehydes I Fehling's is a mild oxidising agent unqualified I aldehydes are easier to oxidise than alcohols I compound A has no aldehyde functional groups. I reacts with Fehling's I carbonyl R ketones are oxidised by Fehling's
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2. M/J 26/P22/Q6/c(iii)

6(c)(iii)	M1 Br₂ (aq) / bromine water M2 orange / brown to colourless in S (not R) OR M1 Na₂CO₃ / sodium carbonate / NaHCO₃ / sodium hydrogencarbonate M2 fizzing / effervescence in R (not S)	2 R capitalisation once e.g. KMNO₄ M2 observation dependent on unambiguous attempt at M1 reagent e.g. M1 Br × M2 decolourises S ✓ OR M1 K₂Cr₂O₇ / Cr₂O₇²⁻ (no acid) ^ M2 (orange) to green ✓ OR M1 NaCO₃ × M2 bubbles in R ✓ etc M2 I alkene for S / carboxylic acid for R but A alcohol for S M1 A bromine / Br₂ M1 A aqueous iodine / aqueous I₂ / iodine water NOT iodine alone M1 R iodoform M2 A decolourises M2 A yellow to colourless M2 R red M2 A bubbles M2 I limewater (turns milky)
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6(c)(iii)		Alternatives: M1 acidified potassium dichromate / H^+ AND $\text{K}_2\text{Cr}_2\text{O}_7$ M1 BOD H^+ / $\text{Cr}_2\text{O}_7^{2-}$ M2 (orange) to green with S (not R) M1 potassium permanganate / KMnO_4 M1 BOD potassium manganate M1 BOD MnO_4^- M2 decolorises / (purple) to colourless (solution) / brown ppt with S (not R) M2 A green solution formed if alkaline KMnO_4 used M1 addition of named alcohol and (conc) sulfuric acid OR named alcohol and catalyst M2 sweet smell with R (not S) M1 addition of named carboxylic acid and (conc) sulfuric acid OR named carboxylic acid AND catalyst M2 sweet smell with S (and not R) M1 named acid-base indicator M2 appropriate colour change with R (and not S) M1 A Mg / magnesium M2 A faster bubbles with R (comparison required)
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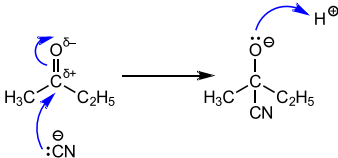
3. M/J 26/P23/Q4/a(iii)

4(a)(iii)	Alcohols / Compound V are / is not oxidised by Fehling's reagent (aldehydes are oxidised by Fehling's reagent) OR Aldehydes are oxidised by Fehling's reagent (alcohols / Compound V are / is not oxidised by Fehling's reagent)	1 A for alcohol groups = hydroxy = hydroxyl = OH group R for alcohol groups = hydroxide / OH^- group A alcohol groups are not a strong enough reducing agent (to reduce Fehling's) A Fehling's reagent is not a strong enough oxidising agent to oxidise the alcohol groups A Fehling's is a strong enough oxidising agent to oxidise aldehydes A Fehling's is too mild to oxidise alcohols A Fehling's is not reduced by V / alcohols A Fehling's is reduced by aldehydes I Fehling's is a mild oxidising agent (unqualified) I aldehydes are easier to oxidise than alcohols I compound V has no aldehyde functional groups. I reacts with Fehling's I carbonyl R ketones are oxidised by Fehling's
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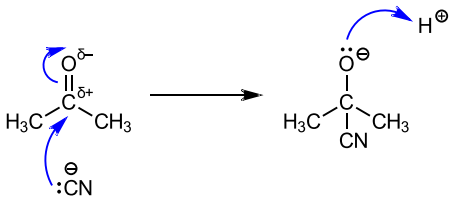
4. M/J 25/P21/Q6/a,b,c(i,ii)

6(a)	carbonyl	1
6(b)	M1 (2,4)-dinitrophenylhydrazine / (2,4)-DNPH M2 orange / red / yellow precipitate	2
6(c)(i)	F = propanone G = ethanal H = propanal	2
6(c)(ii)	tri-iodomethane OR CHI_3	1

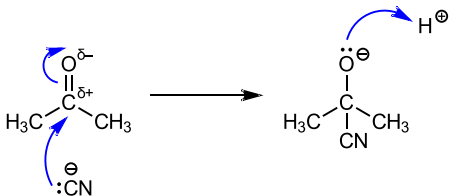
5. MAR 25/P22/Q4/b(i)

4(b)(i)	 • curly arrow from lone pair on :CN to C(δ^+) • curly arrow from C=O to O(δ^-) AND correct dipole • correct intermediate • curly arrow from lone pair on O:$^-$ to H$^+$	4
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6. O/N 24/P21/Q4/a(i),c,d,e

4(a)(i)	 • curly arrow from lone pair on C of CN$^-$ to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H$^+$	1
		1
		1
4(c)(i)	$\text{CH}_3\text{COCH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	1
4(c)(ii)	propan-2-ol	1
4(d)	red / orange / yellow ppt	1
4(e)	Fehling's solution cannot oxidise ketones OR ketones are not easily oxidised	1

7. O/N 24/P23/Q4/a(i),c,d,e

4(a)(i)	 • curly arrow from lone pair on C of CN$^-$ to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H$^+$	1
		1
		1

4(c)(i)	$\text{CH}_3\text{COCH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	1
4(c)(ii)	propan-2-ol	1
4(d)	red / orange / yellow ppt	1
4(e)	Fehling's solution cannot oxidise ketones OR ketones are not easily oxidised	1

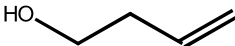
8. M/J 24/P21/Q6/b(i),c(ii)

6(b)(i)	reducing agent for C=O AND C=C		1
6(c)(ii)		observation on addition of 2,4-DNPH	1
	Y	orange precipitate	
	Q	no precipitate	
	Both correct for one mark		

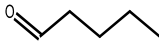
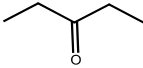
9. MAR 24/P22/Q1/d(iii,iv)

1(d)(iii)	<table border="1"> <tr> <th>reagent</th><th>observation with Y</th><th>observation with Z</th></tr> <tr> <td>$\text{Na}_2\text{CO}_3(\text{aq})$</td><td>no reaction</td><td>• no reaction</td></tr> <tr> <td>alkaline $\text{I}_2(\text{aq})$</td><td>• yellow precipitate (ppt)</td><td>• yellow ppt</td></tr> <tr> <td>2,4-DNPH</td><td>• red / orange / yellow ppt</td><td>• red / orange / yellow ppt</td></tr> <tr> <td>Tollens' reagent</td><td>• no reaction</td><td>• silver mirror</td></tr> </table>	reagent	observation with Y	observation with Z	$\text{Na}_2\text{CO}_3(\text{aq})$	no reaction	• no reaction	alkaline $\text{I}_2(\text{aq})$	• yellow precipitate (ppt)	• yellow ppt	2,4-DNPH	• red / orange / yellow ppt	• red / orange / yellow ppt	Tollens' reagent	• no reaction	• silver mirror	4
reagent	observation with Y	observation with Z															
$\text{Na}_2\text{CO}_3(\text{aq})$	no reaction	• no reaction															
alkaline $\text{I}_2(\text{aq})$	• yellow precipitate (ppt)	• yellow ppt															
2,4-DNPH	• red / orange / yellow ppt	• red / orange / yellow ppt															
Tollens' reagent	• no reaction	• silver mirror															
1(d)(iv)	$\text{CH}_3\text{CHO} + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{OH}$	1															

10. M/J 23/P21/Q4/a

4(a)(i)		1
4(a)(ii)	reducing agent	1

11. M/J 23/P22/Q5/b

5(b)(i)	carbonyl (group)	1
5(b)(ii)	 A  B	2

12. M/J 23/P23/Q5/b,d

5(b)(i)	$\text{C}_6\text{H}_{10}\text{O}_3 + [6\text{H}] \rightarrow \text{C}_6\text{H}_{14}\text{O}_2 + \text{H}_2\text{O}$	1												
5(b)(ii)	LiAlH_4 (in dry ether)	1												
5(d)	<table border="1"> <tr> <th>reagent</th><th>observation on addition to X</th><th>observation on addition to Y</th></tr> <tr> <td>sodium carbonate solution</td><td>effervescence</td><td>no change</td></tr> <tr> <td>2,4-DNPH</td><td>orange precipitate</td><td>no change</td></tr> <tr> <td>alkaline aqueous iodine</td><td>yellow precipitate</td><td>yellow precipitate</td></tr> </table> Two correct for one mark, four correct for two marks, six correct for three marks	reagent	observation on addition to X	observation on addition to Y	sodium carbonate solution	effervescence	no change	2,4-DNPH	orange precipitate	no change	alkaline aqueous iodine	yellow precipitate	yellow precipitate	3
reagent	observation on addition to X	observation on addition to Y												
sodium carbonate solution	effervescence	no change												
2,4-DNPH	orange precipitate	no change												
alkaline aqueous iodine	yellow precipitate	yellow precipitate												

13. M/J 23/P23/Q6/b(i,ii)

6(b)(i)	nucleophilic addition	1
6(b)(ii)	M1 correct dipole on C=O AND curly arrow from C=O bond to O AND from lone pair of electrons on C of :CN⁻ M2 correct intermediate (charge on O or overall) M3 curly arrow from lone pair of electrons on O to H⁺ AND correct organic product (+ CN⁻)	3

14. MAR 23/P22/Q3/b

3(b)(i)	curly arrow from lone pair on C of CN⁻ to C^(δ+)	1
	correct dipole on C=O AND curly arrow from C=O bond to O ^{δ-}	1
	correct organic intermediate	1
	curly arrow from lone pair on alkoxide O ⁽⁻⁾ to H ⁺ / H of HCN / H of H ₂ O (to reform the alcohol)	1
3(b)(ii)	acidified K ₂ Cr ₂ O ₇	1
	Tollens' reagent	1
	no reaction	1
3(b)(iii)	C ₄ H ₈ O + [O] → C ₄ H ₈ O ₂	1
3(b)(iv)		1

15. O/N 22/P21/Q5/a(ii,iii)

5(a)(ii)	C ₆ H ₁₀ O ₃ + 2[H] → C ₆ H ₁₂ O ₃	1
5(a)(iii)	M1 (ketone in) N is planar (so can be attacked from either side)	1
	M2 because different stereoisomers / optical isomers form	1

16. O/N 22/P23/Q5/a(ii,iii)

5(a)(ii)	C ₆ H ₁₀ O ₃ + 2[H] → C ₆ H ₁₂ O ₃	1
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5(a)(iii)	M1 (ketone in) N is planar (so can be attacked from either side)	1
	M2 because different stereoisomers / optical isomers form	1

17. M/J 22/P21/Q3/c(i)

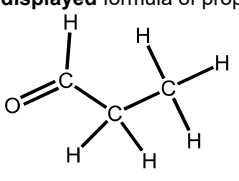
3(c)(i)	NaBH ₄ / sodium borohydride / LiAlH ₄ / lithium tetrahydridoaluminate	1
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18. M/J 22/P21/Q4/c(i,ii)

4(c)(i)	carbonyl	1
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4(c)(ii)	CHI ₃	1
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19. M/J 22/P23/Q3/c

3(c)	displayed formula of propanal  M1 identification of propanal M2 correct use of displayed formula and correct formula used	2
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20. MAR 22/P22/Q4/a(i)

4(a)(i)	red / orange / yellow precipitate / ppt / solid [1] silver mirror / silver / grey solid / precipitate / ppt [1] effervescence / bubbling / fizzing [1]	3
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21. O/N 21/P22/Q3/d(ii),e(i,iii)

3(d)(ii)	M1 (2,4-DNPH will produce a) red / orange / yellow precipitate M2 V has a carbonyl group	2
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3(e)(i)	yellow precipitate	1
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3(e)(iii)	$C_6H_{12}O + 2[H] \rightarrow C_6H_{14}O$	1
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22. M/J 21/P23/Q4/a(ii,iii)

4(a)(ii)	carbonyl / ketone	1
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4(a)(iii)	NaBH ₄ OR LiAlH ₄	1
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23. MAR 21/P22/Q4/b,d(i,ii)

4(b)	M1: red / orange / yellow ppt / solid M2: silver mirror OR silver / grey / black / brown ppt / solid	2
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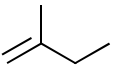
4(d)(i)	$CH_2OHCHO + 2[H] \rightarrow (CH_2OH)_2$	1
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4(d)(ii)	NaBH ₄ / LiAlH ₄	1
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24. O/N 20/P21/Q4/c

4(c)(i)	2(-)iodo(-)2(-)methylbutane	1
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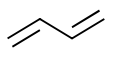
4(c)(ii)	Nucleophilic substitution / S _N	1
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4(c)(iii)		1
4(c)(iv)	(L has) two identical / two methyl groups attached to one end / one carbon of the C=C / double bond	1
4(c)(v)	ethanoic acid / CH ₃ COOH	1
4(c)(vi)	CH ₃ COCH ₃ + 3I ₂ + 4OH ⁻ → (1)CH ₃ COO ⁻ + 3H ₂ O + 3I ⁻ + CHI ₃ M1: correctly balanced M2: CHI ₃ product	2
4(c)(vii)	yellow ppt / yellow solid	1

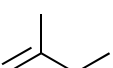
25. O/N 20/P22/Q3/c(i)

3(c)(i)	LiAlH ₄ OR lithium tetrahydridoaluminate((III))	1
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26. O/N 20/P22/Q4/a(iv),b(ii,iii,iv)

4(a)(iv)	(formation of) silver mirror / ppt	1
4(b)(ii)	M1: add aqueous alkaline iodine M2: G no change AND J yellow ppt	2
4(b)(iii)	reducing agent	1
4(b)(iv)	(1,3-)butadiene OR buta(-1,3-)diene OR 	1

27. O/N 20/P23/Q4/c

4(c)(i)	2(-)iodo(-)2(-)methylbutane	1
4(c)(ii)	Nucleophilic substitution / S _N	1
4(c)(iii)		1
4(c)(iv)	(L has) two identical / two methyl groups attached to one end / one carbon of the C=C / double bond	1
4(c)(v)	ethanoic acid / CH ₃ COOH	1
4(c)(vi)	CH ₃ COCH ₃ + 3I ₂ + 4OH ⁻ → (1)CH ₃ COO ⁻ + 3H ₂ O + 3I ⁻ + CHI ₃ M1: correctly balanced M2: CHI ₃ product	2
4(c)(vii)	yellow ppt / yellow solid	1

28. M/J 20/P21/Q5/c(i)

5(c)(i)	carbonyl / aldehyde / ketone	1
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29. M/J 20/P21/Q6/d

6(d)	M1 reagent (2,4-) DNPH / (2,4)-dinitrophenylhydrazine M2 observation yellow / orange / red precipitate	2
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30. M/J 20/P22/Q5/b(i)

5(b)(i)	<i>mechanism for ethanal and HCN using CN⁻ (from KCN) as the catalyst</i> M1 arrow from lone pair of electrons on :CN⁻ to C of C=O M2 correct dipole on C^{δ+}=O^{δ-} AND arrow from the double bond to or beyond the O of C=O M3 arrow from lone pair of electrons on O of intermediate to H of HCN AND arrow from H-C bond to C of H-C≡N	3
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31. MAR 20/P22/Q3/a(ii,iii)

3(a)(ii)	nucleophilic addition	1
3(a)(iii)	yellow / orange / red ppt / solid	1

32. O/N 19/P21/Q4/b(i,iii),c(i,ii,vii,viii)

4(b)(i)	acidified potassium dichromate[[VI]] AND heat under reflux	1
4(b)(iii)	$(\text{CH}_3)_3\text{CCHO} + 2[\text{H}] \rightarrow (\text{CH}_3)_3\text{CCH}_2\text{OH}$	1
4(c)(i)	- orange / red / yellow precipitate - orange / red / yellow precipitate	1
4(c)(ii)	Aldehyde	1
4(c)(vii)	M1: Correct structure of X and correct dipole on C=O M2: curly arrow from C=O bond to O AND intermediate with CN attached and -ve charge on the O M3: curly arrow from lone pair on CN⁻ to C(=O) in X AND curly arrow from lone pair in the intermediate to H⁺	3
4(c)(viii)	catalyst	1

33. O/N 19/P23/Q4/b(i,iii),c(i,ii,vii,viii)

4(b)(i)	acidified potassium dichromate[[VI]] AND heat under reflux	1
4(b)(iii)	$(\text{CH}_3)_3\text{CCHO} + 2[\text{H}] \rightarrow (\text{CH}_3)_3\text{CCH}_2\text{OH}$	1
4(c)(i)	- orange / red / yellow precipitate - orange / red / yellow precipitate	1
4(c)(ii)	Aldehyde	1

4(c)(vii)	M1: Correct structure of X and correct dipole on C=O M2: curly arrow from C=O bond to O AND intermediate with CN attached and –ve charge on the O M3: curly arrow from lone pair on CN[–] to C(=O) in X AND curly arrow from lone pair in the intermediate to H⁺	3
4(c)(viii)	catalyst	1

34. M/J 19/P21/Q4/d(i)

4(d)(i)	A and B	1
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35. M/J 19/P21/Q5/a

5(a)	M1 a lone pair / electron pair donor	1
	M2 (:)CN [–] / [–] (:)CN / cyanide ion	1

36. MAR 19/P22/Q3/b

3(b)	C ₅ H ₁₀ O + 2[H] → C ₅ H ₁₂ O	1
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37. O/N 18/P21/Q4/c

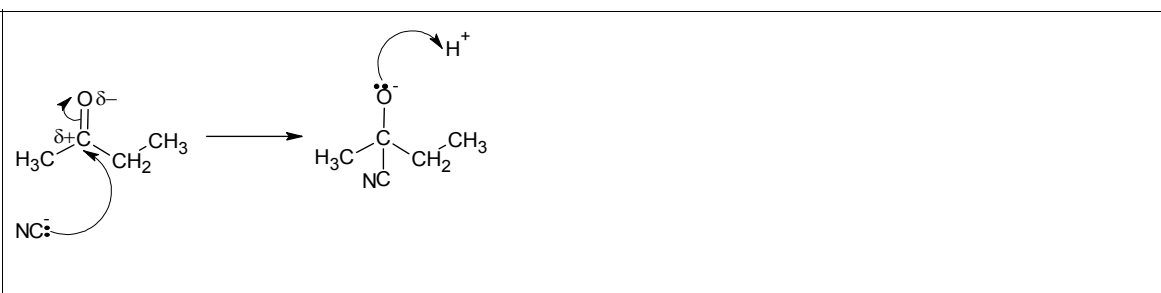
4(c)(i)	reducing agent / reductant	1
4(c)(ii)	C ₂ H ₂ O ₃ + 2[H] → C ₂ H ₄ O ₃ M1 for correct molecular formulae C ₂ H ₂ O ₃ and C ₂ H ₄ O ₃ M2 for balancing	2

38. O/N 18/P23/Q4/c

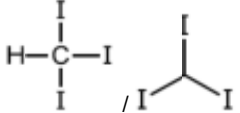
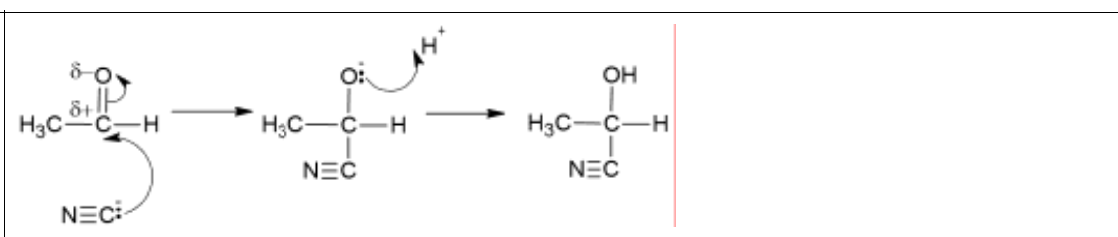
4(c)(i)	reducing agent / reductant	1
4(c)(ii)	C ₂ H ₂ O ₃ + 2[H] → C ₂ H ₄ O ₃ M1 for correct molecular formulae C ₂ H ₂ O ₃ and C ₂ H ₄ O ₃ M2 for balancing	2

39. M/J 18/P22/Q4/a(i),c,d(i)

4(a)(i)	iodoform / tri-iodomethane	1
4(c)(i)	reduction	1
4(c)(ii)	NaBH ₄ / sodium borohydride OR LiAlH ₄ / lithium aluminium hydride	1

4(d)(i)		
	lone pair on C of CN ⁻ and curly arrow to C of C=O	1
	correct dipole on C=O and curly arrow from = to O	1
	correct intermediate anion	1
	curly arrow from lone pair on O to H ⁺	1

40. M/J 17/P21/Q4/c,d(ii)

4(c)(i)	P = propanal	1
	Q = propanone	1
4(c)(ii)	 tr(i)iodomethane / CHI ₃ /	1
4(d)(ii)		
	curly arrow from lone pair on :C≡N to C ^(δ+)	1
	correct dipole on carbonyl δ ⁺ C=O ^{δ-} AND curly arrow from bond to O ^(δ-)	1
	correct intermediate, including C—O ⁻ AND curly arrow from lone pair to H ⁺	1

41. M/J 17/P22/Q4/c(i,ii)

4(c)(i)	triiodomethane	1
4(c)(ii)	F = CH ₃ CH ₂ CH ₂ COCH ₃	1
	G = C ₂ H ₅ CH(CH ₃)CHO	1

42. O/N 16/P21/Q5/d

5(d)	(1-)propyl propanoate	1 1
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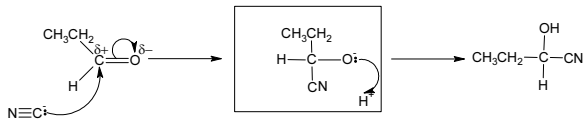
43. O/N 16/P22/Q5/b

5(b)(i)	Oxidation OR reduction	1
5(b)(ii)	Condensation	1
5(b)(iii)	Reduction OR oxidation OR addition	1
5(b)(iv)	Addition	1

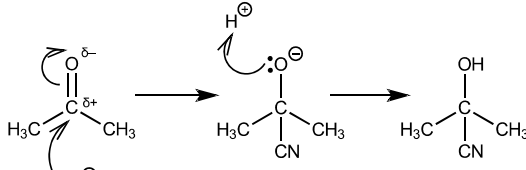
44. O/N 16/P23/Q5/d

5(d)	(1-)propyl propanoate	1	1
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45. M/J 16/P22/Q5/c(i,iii)

(c) (i)	 M1 = lone pair on C of CN⁻ AND curly arrow from lone pair to carbonyl carbon M2 = dipole on C=O AND curly arrow to O from = M3 = intermediate with negative charge M4 = lone pair and curly arrow to H⁺	[1] [1] [1] [1]	[4]
(iii)	attack/attach from either side/above or below/from two directions because the carbonyl/ molecule is planar/trigonal/flat/because of the shape of the molecule OR product is chiral/has a chiral carbon/has a carbon attached to four different groups/has a chiral centre/is asymmetric (equal) chance of forming either (of the two optical isomers)/mechanism doesn't distinguish between the two (optical isomers)/able to form either/chance of forming/able to form 50:50 OR because the carbonyl/ molecule is planar/trigonal/flat OR because of the shape of the molecule (equal) chance of forming either (of the two optical isomers)/mechanism doesn't distinguish between the two (optical isomers)/able to form either/chance of forming/able to form 50:50	[1] [1]	[2]

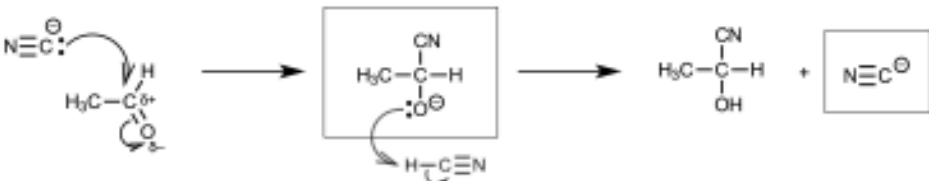
46. MAR 16/P22/Q3/d

(d) (i)	nucleophilic addition	[1]	[1]
(ii)	 correct dipole on carbonyl curly arrow from lone pair on CN⁻ AND from C=O to O correct intermediate curly arrow from lone pair on O⁻ to H⁺ correct product	[1] [1] [1] [1] [1]	[5]

47. MAR 16/P22/Q5/a(iii)

(iii)	red / orange / yellow precipitate / solid	[1]	[1]
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48. M/J 15/P21/Q3/b(ii),c

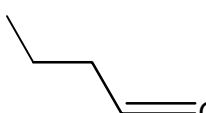
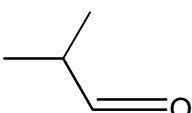
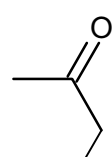
(ii)	(Planar) carbonyl so (equal chance of nucleophile) attacking either side	[1]	[1]
3 (c) (i)	 M1 = lone pair AND curly arrow from lone pair to carbonyl C M2 = partial charges on C=O AND curly arrow from bond (=) to O^{δ-} M3 = structure of intermediate including charge M4 = lone pair AND two correct curly arrows (from lone pair to H AND from H—C to C) M5 = CN⁻	[1] [1] [1] [1] [1]	[5]

(ii)	(CN ⁻ regenerated so) catalyst	[1]	[1]
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49. M/J 15/P22/Q4/b

(b) (i)	Fehling's / Benedict's OR Tollens' OR dichromate OR manganate Warm / heat	[1]	
	Fehling's / Benedict's = (Brick)-red ppt Tollens' = silver / mirror OR grey / black precipitate Dichromate = orange to green Manganate = purple to colourless } with the aldehyde / A-D	[1]	
(ii)	(2,4-)DNP(H) / Brady's reagent	[1]	
	Orange / yellow / red-orange / yellow-orange ppt	[1]	

50. M/J 15/P23/Q3/a(i,iii),c

3	(a)	(i)	A/B =  	[1]			
				[1]			
		C = 	[1]				
					[3]		
(iii)				Silver mirror/ppt/solid (black/grey)	[1]	[1]	
(c)				(i)	$C_3H_6O + [O] \rightarrow C_3H_6O_2$	[1]	[1]
				(ii)	$C_3H_6O + 2[H] \rightarrow C_3H_8O$	[1]	[1]